
From Adversarial Robustness to Interpretability: Toward Quantitative Metrics for Transparent Artificial Intelligence

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Abstract

Recent work on evaluating mechanistic interpretability has yielded important qualitative insights into the internal representations and computations of neural networks, yet it lacks the standardized quantitative metrics and benchmarks that have enabled adversarial robustness to develop into a reproducible, comparative science. This disparity motivates a fundamental question: can mechanistic interpretability become quantitatively measurable in the same way that adversarial robustness has? We investigate this question by proposing feature superposition as a common geometric variable linking interpretability and adversarial robustness. Our central hypothesis is that lower feature superposition, corresponding to more monosemantic representations, is associated with both greater interpretability and greater adversarial robustness.

We make three contributions. First, we formulate a testable geometric account connecting feature superposition, representational structure, interpretability, and adversarial vulnerability. Second, we develop a representation-based approach for quantitatively characterizing this structure, providing a basis for comparing interpretability across neural networks rather than relying solely on architecture-specific qualitative analysis. Third, we empirically examine the proposed relationship across architectures and datasets. The findings provide empirical support for the proposed connection between structured, less superposed representations and interpretability, while moti-

vating further investigation of their relationship to adversarial robustness.

Our results suggest that InterpScore, the proposed quantitative interpretability metric, is sensitive to controlled representational capacity, yet weakly correlated with conventional performance/robustness measures. These findings indicate that successful adversarial perturbations can provide a measurable bridge between two previously more separate research directions and a foundation for developing standardized quantitative evaluation in mechanistic interpretability. More broadly, the work advances a shift from predominantly qualitative descriptions of internal mechanisms toward measurable, reproducible, and cross-model analysis of neural representations, analogous to the role that quantitative evaluation played in the maturation of adversarial robustness.

1 INTRODUCTION

Adversarial robustness offers an instructive example of how a relatively qualitative research problem can develop into a quantitative scientific discipline. Over the past decade, the study of adversarial vulnerability has progressed from the discovery of adversarial examples Szegedy et al. (2013) to increasingly standardized threat models, attack procedures, and evaluation protocols. Perturbation budgets such as ϵ provide a common language for defining threats, attacks such as PGD and AutoAttack provide increasingly reliable procedures for finding adversarial examples Madry et al. (2018); Croce and Hein (2020), and benchmarks such as RobustBench provide standardized evaluations and public leaderboards across datasets and architectures Croce et al. (2020). As a result, robustness can be compared quantitatively across substantially different models using common metrics such as robust accuracy and certified perturbation radii. The importance of this infrastructure is not merely organi-

zational: standardized measurements make results reproducible, enable comparisons across studies, and allow progress to accumulate across the research community.

Mechanistic interpretability has developed along a rather different trajectory. The field has produced a growing body of detailed qualitative and mechanistic explanations of neural networks, including induction heads in transformers Elhage et al. (2021), curve-detection circuits in vision models Cammarata et al. (2020), and circuit-level analyses of language models. These studies demonstrate that seemingly opaque neural networks can contain identifiable computational structures whose functions can be described in human-understandable terms. At the same time, quantitative work has begun to measure particular aspects of interpretability, including semantic alignment, concept sensitivity, and automated behavioral consistency Bau et al. (2017); Kim et al. (2018); Zimmermann et al. (2024). Nevertheless, unlike adversarial robustness, mechanistic interpretability does not yet have a single widely adopted quantitative quantity that captures an important structural property of representations and can be used systematically to compare models, layers, and architectures. Consequently, statements about one model being “more interpretable” than another often remain dependent on the particular circuit, feature, task, or qualitative criterion under investigation.

This difference creates a measurement problem. Without quantitative measurements that can be reproduced across models and experimental settings, interpretability results can remain local: individual circuits can be carefully understood without necessarily yielding a common scale on which broader architectural or training effects can be compared. The central premise of this work is that mechanistic interpretability can benefit from the same methodological shift that proved valuable in robustness: identify a meaningful structural property, define a reproducible quantitative measurement of that property, and evaluate it under standardized experimental conditions. The goal is not to reduce interpretability to a single universal scalar, but to make an important aspect of representation structure measurable.

We focus on *feature superposition* as a candidate for such a structural quantity. Superposition describes the regime in which a network represents more features than it has available representational dimensions, causing multiple features to share directions in activation space and potentially producing polysemantic neurons. Toy models have demonstrated that this phenomenon can arise naturally when sparse features compete for limited representational capacity, and have further shown that superposition has a rich geometric

structure and may be connected to adversarial examples Elhage et al. (2022). This perspective suggests a possible connection between two properties that are often studied largely independently: the interpretability of internal representations and their vulnerability to adversarial perturbations.

Our hypothesis is that these properties are related through representation geometry. When feature representations concentrate in a small number of directions, the resulting activation geometry is relatively low-dimensional and may admit simpler mechanistic descriptions. Conversely, when features are encoded in overlapping directions, representational variation becomes more distributed and polysemantic. We use adversarial perturbations as a natural probe of this geometry: rather than examining arbitrary points in representation space, we study representations reached by perturbations that successfully change the model’s prediction. Under this hypothesis, the geometry of successful adversarial representations should contain information about feature superposition and can thus be quantified as a representation-level measure of interpretability.

We make four contributions. First, we formulate a geometric account of interpretability linking feature superposition, representation geometry, and adversarial vulnerability through successful adversarial representations. Second, we introduce a quantitative measure that characterizes this structure using geometric, overlap, and separation descriptors of the adversarial representation manifold, without requiring semantic annotations, a predefined feature vocabulary, a probe architecture, or explicit feature decomposition. Third, we establish empirically testable invariance and comparability properties, enabling consistent evaluation across layers, model widths, architectures, datasets, and training regimes. Fourth, we validate the approach on controlled synthetic data with known ground truth and on real-world vision datasets (CIFAR-10, CUB200, and TinyImageNet-200), using ResNet and ViT architectures to test whether the measure distinguishes standard from adversarially trained representations and captures systematic variation with layer depth and model width.

Together, these experiments show how a scalar geometric measure can compare representations across models and training regimes under a common evaluation procedure.

2 RELATED WORK

Our work is situated within efforts to make interpretability quantitative and experimentally testable. Recent work has developed objective measures of se-

mantic alignment, feature attribution, representation similarity, and internal structure. We organize the literature into four themes: quantitative interpretability and XAI; representation geometry and feature superposition; adversarial perturbations and mechanistic analysis; and our methodological position.

2.1 Quantitative Interpretability and XAI

A major line of work quantifies interpretability through semantic concepts or feature attribution. Network Dissection measures alignment between neural units and human-interpretable visual concepts Bau et al. (2017), while TCAV quantifies model sensitivity to user-defined concepts Kim et al. (2018). These approaches provide quantitative measures of semantic alignment, but rely on externally specified concept vocabularies and primarily evaluate individual units or concept directions.

Explainable AI has also developed attribution and perturbation-based methods for identifying features that influence predictions. Saliency maps Simonyan et al. (2014), Integrated Gradients Sundararajan et al. (2017), DeepLIFT Shrikumar et al. (2017), and meaningful perturbation Fong and Vedaldi (2017) provide different attribution approaches, while feature visualization and activation analysis examine internal representations directly ?? . Importantly, Adebayo et al. (2018) showed that attribution methods can exhibit properties that complicate their interpretation, motivating controlled quantitative evaluation. Automated metrics such as the Machine Interpretability Score (MIS) have further sought scalable measures without exhaustive human annotation Zimmermann et al. (2024). These methods quantify semantic alignment, attribution, or explanation quality rather than the geometry of the representation itself.

2.2 Representation Geometry and Feature Superposition

A complementary perspective treats neural representations as geometric objects rather than collections of independently interpretable units. Distributed representations encode information across many dimensions ?Bengio et al. (2013), and subsequent work has developed methods for comparing learned representation geometry Morcos et al. (2018); Kornblith et al. (2019). This perspective is central to the feature superposition hypothesis: Elhage et al. (2022) showed in toy models that networks with fewer dimensions than potentially useful features can represent multiple features using overlapping directions in activation space, providing a mechanistic account of polysemanticity. Subsequent work has investigated recovering such features using

sparse coding and sparse autoencoders Sharkey et al. (2022); Bricken et al. (2023); Huben et al. (2024); ?. These approaches provide powerful tools for recovering latent features, but typically assume an explicit feature-recovery procedure or model of latent structure. Our approach instead asks whether naturally occurring model responses to successful adversarial perturbations can themselves provide a reproducible probe of this structure.

2.3 Adversarial Perturbations and Mechanistic Analysis

Adversarial examples provide a complementary means of probing neural representations. Small perturbations can systematically alter predictions Szegedy et al. (2014); Goodfellow et al. (2015), while analyses of robust and non-robust features suggest that imperceptible perturbations can reveal information encoded by a model that is not apparent from human-perceptible semantics Ilyas et al. (2019); ?. Adversarial training provides a framework for controlling this behavior Madry et al. (2018). Perturbations have also been used for interpretation: meaningful perturbation identifies input regions whose modification changes model outputs Fong and Vedaldi (2017), while activation-based approaches and feature visualization examine internal responses ?? . These methods motivate perturbations as controlled probes, but generally focus on attribution or individual predictions rather than the geometry of resulting representations.

Mechanistic interpretability similarly seeks structural descriptions of computation through circuits and mechanistic analysis Olah et al. (2020); Elhage et al. (2021); Wang et al. (2022), with causal interventions testing the role of internal components Geiger et al. (2021); Meng et al. (2022). These approaches emphasize internal structure and causality but do not directly provide a general quantitative measure of feature superposition in realistic representations.

2.4 Related Work and Methodological Position

Taken together, these literatures address complementary aspects of interpretability: semantic alignment and concept sensitivity Bau et al. (2017); Kim et al. (2018); attribution and explanation Simonyan et al. (2014); Shrikumar et al. (2017); Sundararajan et al. (2017); scalable interpretability metrics Zimmermann et al. (2024); and feature superposition and recovery Elhage et al. (2022); Sharkey et al. (2022); Bricken et al. (2023). Our methodological position differs in focusing directly on representation geometry. Rather than assigning semantic labels, specifying a concept

vocabulary, recovering an explicit sparse feature dictionary, or explaining an individual prediction, we use successful adversarial perturbations as a reproducible probe of the representation. The resulting activations define a geometric structure whose covariance, overlap, and separation can be quantified across models, layers, datasets, and training regimes. This connects adversarial robustness to feature superposition: if multiple features occupy overlapping directions, systematic perturbations that produce behavioral changes may reveal structure in those directions. Our goal is complementary to semantic and feature-recovery methods, providing a representation-level measurement that requires neither a predefined semantic vocabulary nor an explicit latent feature decomposition. More broadly, we address a measurement gap between quantitative interpretability methods focused on semantic alignment or explanation quality and superposition work that studies feature geometry through controlled settings or explicit feature recovery. We investigate whether successful adversarial perturbations can bridge these perspectives by providing a reproducible, model-derived probe of representation geometry in realistic vision models.

3 METHODOLOGY

Let a trained classifier be

$$f : \mathcal{X} \rightarrow \mathcal{Y}, \quad (1)$$

where $\mathcal{X}$ denotes the input space and

$$\mathcal{Y} = \{1, \dots, C\} \quad (2)$$

denotes the set of C output classes.

Given a dataset

$$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N, \quad (3)$$

the objective is to quantify the geometric organization of the learned internal representations as a proxy for interpretability, without requiring a predefined semantic vocabulary, human concept annotations, or a separately trained probing model.

For each analyzed layer $l \in \{1, \dots, L\}$, let

$$\phi_l : \mathcal{X} \rightarrow \mathbb{R}^{d_l} \quad (4)$$

denote the corresponding learned representation.

The proposed measurement is based on successful adversarial transitions. For an input x_n with true class

$$c_t = y_n, \quad (5)$$

an adversarial procedure produces

$$x_n^{\text{adv}} = \mathcal{A}(x_n, y_n). \quad (6)$$

When the attack succeeds,

$$f(x_n) = c_t, \quad f(x_n^{\text{adv}}) = c_p, \quad c_p \neq c_t. \quad (7)$$

The ordered pair

$$(c_t, c_p) \quad (8)$$

therefore defines a successful adversarial transition from the true class to the adversarially predicted class.

The benchmark analyzes

$$\phi_l(x_n^{\text{adv}}) \quad (9)$$

conditional on this transition. Rather than assigning a single global geometric object to each layer, it constructs a collection of transition-specific representation regions and quantifies their within-region geometry and pairwise relationships.

The resulting layer-wise measurement is

$$\mathcal{I}_l = \mathcal{F}(G_l, O_l, S_l; \alpha, Q), \quad (10)$$

where G_l describes within-manifold geometry, O_l measures between-manifold overlap, S_l measures between-manifold separation, and α and Q control the contribution of the corresponding interaction terms.

A network-level summary is subsequently obtained by integrating the layer-wise profile:

$$\mathcal{I}_{\text{int}} = \int_0^1 \mathcal{I}(t) dt. \quad (11)$$

A complementary Target Transition Matrix (TTM) is also constructed from class-conditioned geometric descriptors.

The benchmark consists of a two-pass representation-analysis pipeline followed by geometric scoring and depth aggregation.

The complete pipeline consists of:

1. adversarial representation probing;
2. intermediate representation extraction;
3. dimensionality reduction;
4. construction of transition-specific representation reservoirs;
5. convex-hull approximation;
6. within-manifold geometric measurement;
7. between-manifold geometric measurement;
8. layer-wise score construction;
9. depth-wise integration; and
10. construction of the complementary Target Transition Matrix.

Algorithm 1: Streaming PCA with Convex Hull Tracking

Streams mini-batches through an incremental PCA model, projects them into PCA space, and maintains convex hulls of prediction-target transition manifolds to compute interpretability.

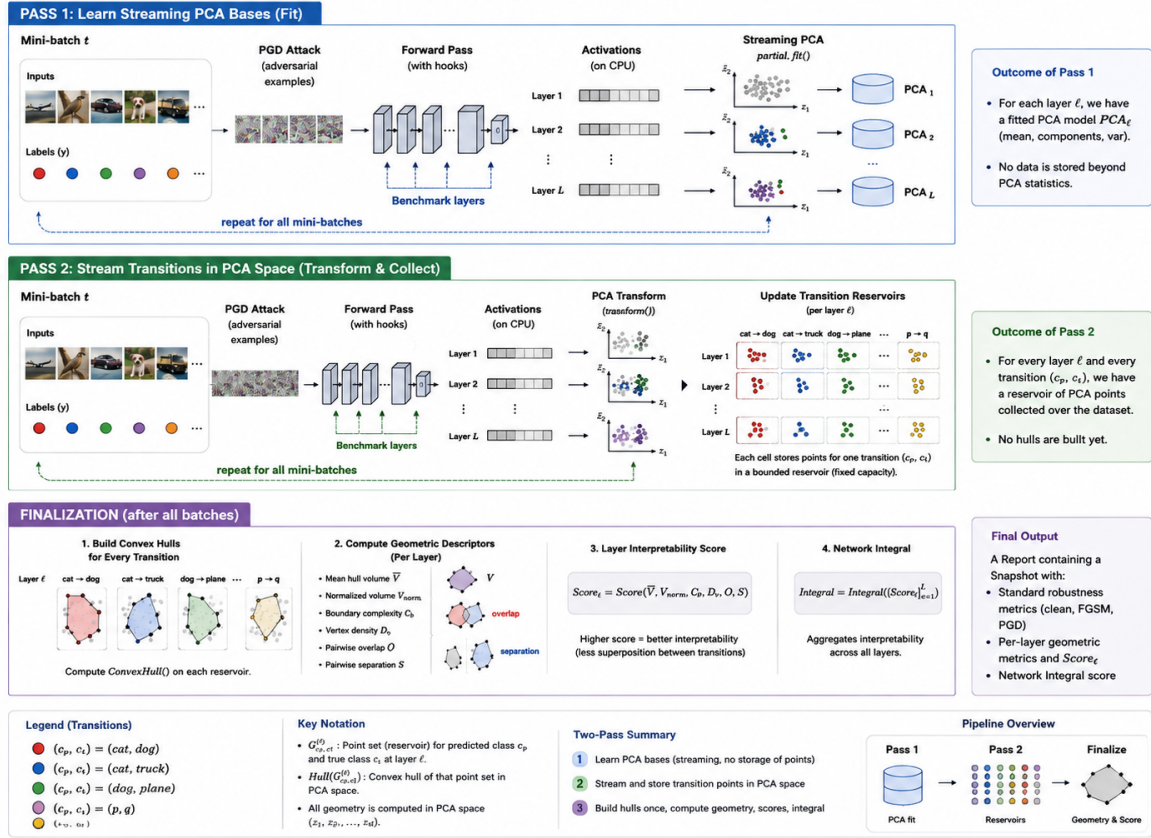


Figure 1: Overview of the proposed benchmark. A trained classifier is first probed using successful adversarial perturbations. Intermediate representations are extracted and reduced to a common low-dimensional space using streaming PCA. The resulting representations are assigned to ordered true-class-to-predicted-class transition reservoirs. Convex hulls approximate the corresponding transition manifolds, from which within-manifold geometry and between-manifold overlap and separation are measured. These quantities are combined into a layer-wise interpretability score and integrated across representation depth. Class-conditioned geometric contrasts are additionally reported through the Target Transition Matrix.

Adversarial Representation Probing. For each correctly classified example (x_n, y_n) , we generate an adversarial example using projected gradient descent (PGD) under an ℓ_∞ perturbation constraint $|\delta_n| * \infty \leq \epsilon$. Starting from the original input, PGD iteratively takes sign-gradient steps with step size $\alpha * \text{PGD}$ and projects the perturbed input back onto the ϵ -ball around x_n . Only successful attacks are retained, i.e., $f(x_n) = y_n$ and $f(x_n^{\text{adv}}) \neq y_n$. Each retained example is associated with an ordered transition (c_t, c_p) , where $c_t = y_n$ is the true class and $c_p = f(x_n^{\text{adv}})$ is the adversarially predicted class. Thus, adversarial examples are conditioned on the direction of the observed decision-boundary transition rather than pooled into a single adversarial distribution. PGD is used as a standard computational primitive; the transition-

conditioned probing framework is the methodological component of interest.

Representation Extraction and Dimensionality Reduction. For every retained adversarial example, intermediate representations are extracted at the layers under analysis using forward hooks. To enable a common geometric analysis, representations are reduced to vectors: convolutional feature maps are spatially averaged over their spatial dimensions, while transformer representations use the CLS token. Samples producing non-finite representations are discarded without modification.

Because intermediate representations can be high-dimensional, a separate streaming PCA model is fitted for each layer. The first pass estimates the layer-

specific mean and the first k principal directions, optionally after feature standardization. In a second pass, adversarial examples are regenerated and their representations are projected using these fixed transformations. This two-pass procedure keeps the geometric coordinate system fixed while avoiding the memory cost of storing all activations. The dimensionality k is kept constant across layers to make subsequent geometric quantities comparable.

Transition-Specific Representation Manifolds.

For each layer l and ordered transition (c_t, c_p) , projected representations are collected into a bounded transition-specific reservoir. The corresponding empirical representation manifold is approximated by the convex hull of the retained points. The direction $c_t \rightarrow c_p$ is explicitly preserved, so transitions in opposite directions are treated as distinct conditional subsets and may produce different geometric regions. Convex hulls provide a simple bounded geometric representation of the observed samples, while the transition-specific decomposition preserves information that would be lost by constructing a single global adversarial manifold.

4 EXPERIMENTS

We evaluate InterpScore through a sequence of experiments designed to assess its validity, sensitivity, generality, and complementarity with conventional performance measures. The experiments are organized around four research questions, each corresponding directly to a specific empirical claim of the proposed benchmark. First, we validate InterpScore in a controlled synthetic environment with known generative factors and systematically varied bottleneck capacity. Second, we assess the sensitivity of the metric to its weighting parameter α using the same fixed representations, thereby isolating the effect of the metric formulation from variation introduced by model training. Third, we evaluate the metric across multiple architectures, datasets, and training regimes to determine whether it remains measurable and stable beyond the controlled setting. Finally, we examine whether InterpScore captures information that is not reducible to predictive accuracy or adversarial robustness.

The experiments address the following research questions:

RQ1: Ground-truth validity. Does InterpScore respond systematically to controlled changes in representational capacity, and does its behavior correspond to an independent characterization of the learned representation based on known generative factors?

This question tests the primary validity claim of In-

terpScore. The synthetic benchmark provides a controlled setting in which bottleneck dimensionality can be manipulated while the underlying generative factors remain fixed. The corresponding ground-truth analysis is computed independently of InterpScore and is therefore used as an external characterization rather than as an optimization target for the metric.

RQ2: Sensitivity to descriptor weighting. How sensitive are InterpScore values and the resulting model ordering to the weighting parameter α ?

This question evaluates whether the conclusions obtained from InterpScore depend critically on a narrow choice of the relative weighting between the geometric and overlap/separation descriptor families. The ablation is performed on fixed trained representations so that changes in the metric can be attributed exclusively to the scoring formulation.

RQ3: Generality and stability. Does InterpScore remain measurable, and comparatively stable across architectures, datasets, and training regimes?

This question evaluates whether the metric extends beyond the controlled synthetic setting to substantially more complex natural-image models. We examine both convolutional and transformer architectures, multiple datasets, and standard and adversarial training.

RQ4: Complementarity with conventional metrics. Does InterpScore capture representation-level information that is not fully explained by predictive accuracy or adversarial robustness?

This question addresses whether InterpScore provides an additional characterization of learned models rather than simply acting as a transformed measure of task performance or robustness. We therefore examine the empirical relationships among InterpScore, accuracy, and PGD accuracy.

4.1 Evaluation Metrics

The primary metric is InterpScore, together with its network-level integral. For the large-scale experiments, we additionally record predictive accuracy and PGD accuracy. These metrics serve different purposes.

InterpScore measures the representation-level properties defined by the proposed geometric formulation. Accuracy measures task performance, while PGD accuracy measures performance under the specified adversarial evaluation procedure. The latter two metrics are therefore not treated as alternative interpretability measures; rather, they provide conventional performance axes against which the information provided

by InterpScore can be compared.

For the synthetic benchmark, the ground-truth factor-level association measurements provide an independent characterization of the learned bottleneck representation.

4.2 Synthetic Dataset

We construct a procedural image dataset in which each image is generated from six independently specified factors: shape, color, size, horizontal position, vertical position, and rotation. The factors are sampled independently and recorded for every example, providing direct ground-truth annotations of the generative factors.

The prediction task depends only on a subset of these factors. Specifically, the target class is a deterministic function of shape and color, whereas size, horizontal position, vertical position, and rotation act as distractor factors. This separation allows us to distinguish task-relevant structure from additional variation represented in the bottleneck.

The controlled construction provides two properties that are important for metric validation. First, the generative factors are known exactly, allowing an independent characterization of learned representations. Second, the task-relevant factors are deliberately separated from nuisance variation, allowing representation-level measurements to be interpreted without relying exclusively on predictive performance.

4.3 Architectures and Bottleneck Capacity

We evaluate two neural architectures, a convolutional neural network (CNN) and a vision transformer (ViT). For each architecture, we vary the bottleneck dimensionality according to

$$M \in \{2, 4, 6, 10, 16\}.$$

For each architecture and bottleneck dimensionality, we train five independently initialized models using different random seeds while keeping the underlying dataset construction fixed. The synthetic benchmark therefore contains

$$2 \times 5 \times 5 = 50$$

independently trained networks.

The controlled variation in M provides a direct manipulation of representational capacity. Smaller bottlenecks impose stronger competition among the underlying factors, whereas larger bottlenecks provide

additional dimensions in which distinct factors can potentially be represented.

4.4 Independent Ground-Truth Characterization

For every trained model, we independently analyze the learned bottleneck representation using the six known generative factors. For each bottleneck dimension and each factor, we compute an association measure, producing a latent-by-factor association matrix.

Because independently trained neural networks need not learn aligned latent coordinate systems, latent dimensions are not averaged directly across models. Instead, for each factor q and each individual model s , we retain the strongest association across bottleneck dimensions,

$$A_q^{(s)} = \max_{m \in \{1, \dots, M\}} A_{m,q}^{(s)}.$$

The resulting factor-level quantities are then averaged across the five random seeds for each architecture and bottleneck dimensionality.

This procedure makes the ground-truth characterization invariant to permutations of the latent dimensions. It also ensures that the oracle measurements are not used to construct, fit, or tune InterpScore.

4.5 Large-Scale Benchmark

To evaluate the generality of our findings beyond the controlled synthetic setting, we consider model-dataset pairings that span distinct architectural families and substantially different data regimes. ResNet-18 and VGG-16 represent convolutional architectures with different structural properties, while ViT-B/16 provides a qualitatively different transformer-based inductive bias. The selected datasets likewise cover different task characteristics, from the relatively low-resolution and coarse-grained CIFAR-10 to the fine-grained CUB200 and the more diverse, ImageNet-like TinyImageNet-200. Importantly, we do not assume that all architecture-dataset combinations are equally well matched; instead, the selected pairings provide complementary validation settings. ResNet-18 across all three datasets allows us to examine the effect of dataset characteristics while holding the architecture fixed, whereas comparisons between ResNet-18 and VGG-16 on CIFAR-10 probe architectural differences under a common data regime. The inclusion of ViT-B/16 further tests whether the observed behavior extends beyond convolutional inductive biases, while the differing suitability of particular architectures to particular datasets itself provides

a useful test of the robustness and scope of our findings. For each configuration, we evaluate both standard and adversarially trained models. Adversarial training uses projected-gradient-descent (PGD) training. Each configuration is evaluated across two independent runs. The resulting benchmark therefore spans multiple architectures, dataset scales, training procedures, and levels of adversarial robustness.

4.6 RQ1: Ground-Truth Validity on Synthetic Data.

We first validate InterpScore in a controlled synthetic setting in which the underlying factors of variation are known exactly. This experiment provides an independent reference against which the behavior of the metric can be evaluated while allowing representational capacity to be varied systematically. The central experimental manipulation is bottleneck dimensionality,

$$M \in \{2, 4, 6, 10, 16\}.$$

For each value of M , five independently trained CNNs and five independently trained ViTs are evaluated.

The experiment tests two related hypotheses. First, in the capacity-constrained regime, increasing bottleneck dimensionality should be accompanied by an increase in InterpScore if the metric is sensitive to the representational competition induced by the bottleneck. Second, after sufficient task-relevant capacity has been obtained, further increases in dimensionality need not produce proportional increases in InterpScore because the supervised task depends primarily on shape and color. The second hypothesis is intentionally treated as an empirical hypothesis rather than as a mathematical consequence of the metric.

InterpScore as a Function of Bottleneck Capacity. Figure 2 reports InterpScore for the five bottleneck widths, separately for the CNN and ViT architectures. Each point corresponds to an independently trained model, while the lines and error bars summarize the distribution across the five random seeds.

Across both architectures, InterpScore increases as the bottleneck width is increased through the capacity-constrained regime. The scores at the smallest bottleneck widths are lower relative to those obtained at larger bottlenecks.

At the largest bottleneck widths, the increase in InterpScore becomes less pronounced. If the observed plateau is present, it is interpreted cautiously. The prediction target depends only on shape and color, whereas the remaining generative factors introduce additional variation that is not required for the super-

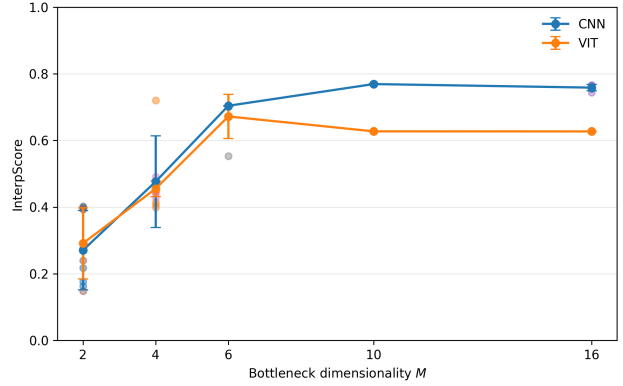


Figure 2: InterpScore as a function of bottleneck dimensionality M for the CNN and ViT architectures. Individual points denote independently trained models at each bottleneck width; lines and error bars summarize the variability across five random seeds.

vised task. Once sufficient capacity is available to represent the task-relevant structure without severe dimensional competition, additional latent dimensions need not necessarily produce a proportional increase in the task-based interpretability score.

The corresponding ground-truth analysis provides an independent test of whether this interpretation is consistent with the actual learned representations.

Ground-Truth Characterization of the Bottleneck. For each trained model, we measure the association between every bottleneck dimension and each of the six ground-truth generative factors. Figure 3 summarizes the resulting factor-level measurements.

The heatmaps provide an independent characterization of how the six known factors are represented as bottleneck capacity changes. In particular, they distinguish the task-relevant factors, shape and color, from the distractor factors, size, position, and rotation.

The resulting patterns partially support the interpretation of the InterpScore curves. Color association increases consistently with bottleneck width (from 0.76 at $M = 2$ to 0.86 at $M = 16$), while shape association remains comparatively stable around 0.26–0.30. The distractor factors likewise show relatively small changes, remaining substantially below the strong color association. Thus, the ground-truth analysis is broadly consistent with increasing availability of task-relevant structure, but does not provide a uniformly monotonic increase across all task-relevant factors and therefore should not be interpreted as a direct one-to-one validation of the InterpScore trend.

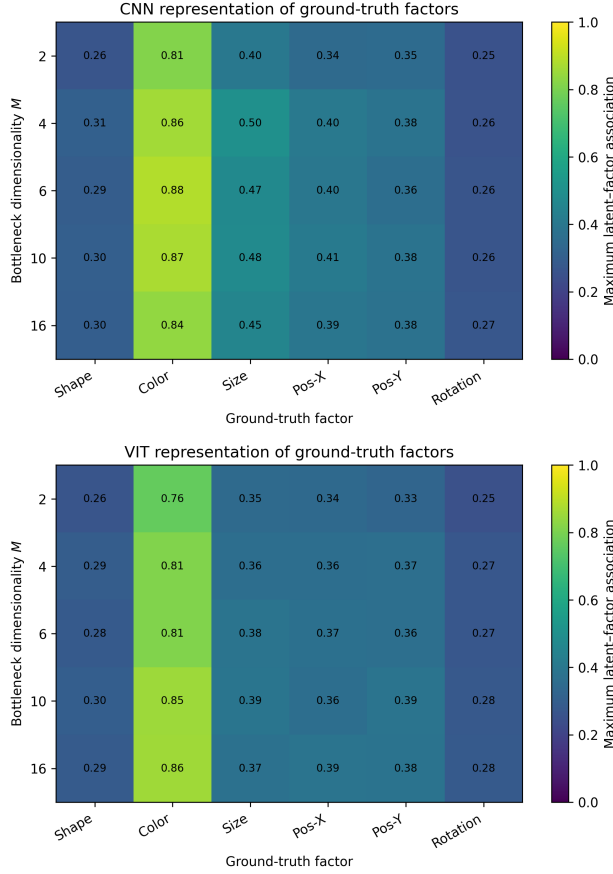


Figure 3: Ground-truth characterization of bottleneck representations. Each cell reports the mean, across five independently trained models, of the strongest association between any bottleneck dimension and the corresponding generative factor. Rows correspond to bottleneck dimensionality and columns to the six generative factors.

Importantly, this ground-truth analysis is not used to define the metric, select its parameters, or construct the score. It is therefore treated as external evidence concerning the behavior of InterpScore.

The ground-truth analysis reveals an additional feature of the learned representations. While color is consistently and strongly associated with the latent representation, shape—the other task-relevant factor—shows comparatively weak association. Interestingly, size exhibits a stronger association than shape across bottleneck widths. This may indicate that the bottleneck preferentially allocates representational capacity to certain factors rather than uniformly preserving the task-generating variables. One possible explanation is that shape and size share correlated or geometrically related structure in the learned representation, such that capacity constraints favor a representation that captures aspects of both through a common

latent direction. However, establishing whether shape and size are genuinely entangled would require an explicit analysis of their representation-level dependence.

Interpretation. Taken together, the synthetic experiment provides two complementary forms of evidence. InterpScore measures the representation using only the quantities defined by the proposed metric, whereas the ground-truth analysis characterizes the same learned representations using factors whose identities are known by construction.

Agreement between these measurements provides evidence that InterpScore responds systematically to independently controlled changes in representational capacity. Disagreement would instead identify a limitation of the metric or of the interpretation assigned to its score.

4.7 RQ2: Sensitivity to the α Weighting Parameter.

We next investigate the sensitivity of InterpScore to its weighting hyperparameter α , which controls the relative contribution of the geometric and overlap/separation descriptor families.

The central purpose of this experiment is to determine whether the qualitative conclusions obtained from InterpScore depend critically on a narrow choice of α . It is not intended to select α by maximizing agreement with the synthetic ground-truth measurements.

Controlled Ablation Design. We use exactly the same trained networks and fixed bottleneck representations as in the preceding synthetic experiment. No model is retrained when α is changed. Consequently, any change in the resulting score is attributable solely to the weighting of the descriptor families.

We evaluate

$$\alpha \in \{0, 0.25, 0.5, 0.75, 1\}.$$

This range spans the two descriptor families from a geometry-dominated formulation to an overlap/separation-dominated formulation.

Figure 4 shows the resulting InterpScore values across bottleneck dimensionalities and values of α . Varying α largely preserves the ordering of bottleneck widths while changing the absolute score. The effect of α is most pronounced at smaller bottleneck widths, whereas the scores become increasingly similar across α as the bottleneck dimension grows.

Overall, the ablation demonstrates that α is a meaningful design parameter of InterpScore rather than

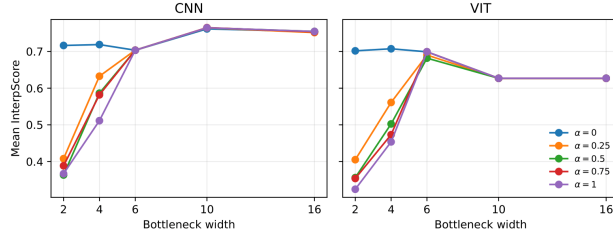


Figure 4: Sensitivity of InterpScore to the descriptor weighting parameter α . Scores are recomputed on the same trained representations for multiple values of α , isolating the effect of the relative weighting between geometric and overlap/separation descriptors. The qualitative dependence on bottleneck dimensionality is largely preserved across values of α , with larger bottlenecks generally yielding higher scores before reaching a plateau.

Table 1: Mean InterpScore across five independently trained runs as a function of the superposition-weighting parameter α . The models and learned representations are held fixed; only α is varied during evaluation.

Architecture	Bottleneck	$\alpha = 0$	$\alpha = .25$	$\alpha = .5$	$\alpha = .75$	$\alpha = 1$
CNN	2	0.717	0.408	0.363	0.388	0.367
	4	0.719	0.633	0.586	0.582	0.511
	6	0.704	0.704	0.704	0.704	0.704
	10	0.762	0.765	0.765	0.765	0.765
	16	0.754	0.752	0.755	0.754	0.755
ViT	2	0.702	0.405	0.356	0.354	0.324
	4	0.708	0.561	0.502	0.473	0.453
	6	0.699	0.690	0.682	0.699	0.699
	10	0.627	0.627	0.627	0.627	0.627
	16	0.627	0.627	0.627	0.627	0.627

a purely numerical scaling constant. Changing the weighting modifies the relative importance assigned to geometric and overlap/separation properties, while preserving the principal qualitative trends of the metric. In particular, the effect of bottleneck dimensionality remains consistent across different values of α , indicating that the observed trends are not driven solely by a particular weighting choice.

We therefore retain

$$\alpha = 0.75$$

as the default setting for the main experiments.

4.8 RQ3: Generality and Stability Across Architectures, Datasets, and Training Regimes.

We next evaluate whether InterpScore remains measurable and informative beyond the controlled syn-

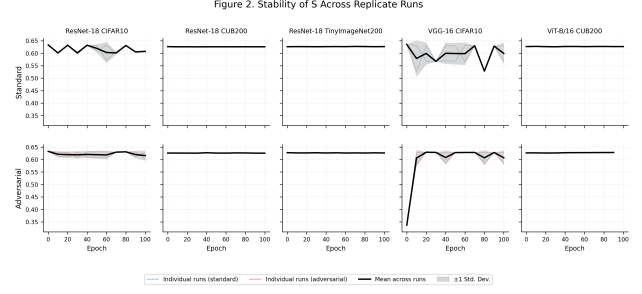


Figure 5: Training dynamics on the large-scale benchmark. InterpScore and InterpScore-Integral are compared with predictive and robustness metrics throughout training across representative model-dataset configurations.

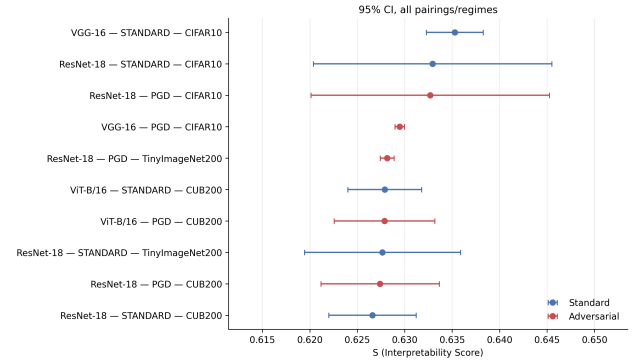


Figure 6: InterpScore and InterpScore-Integral across the large-scale benchmark. Results are shown for ResNet-18 and ViT-B/16 models trained on CIFAR-10, CUB200, and TinyImageNet-200 using standard or adversarial training. Error bars denote the reported uncertainty across independent runs.

thetic setting. We first examine whether InterpScore provides a stable characterization of the learned representation throughout training. Figure 5 shows the evolution of accuracy, PGD accuracy, InterpScore, and InterpScore-Integral over training for the evaluated model-dataset combinations.

InterpScore is largely stable across epochs and replicates for most architecture-datasetpairs; VGG-16 on CIFAR10 shows greater variability, including an initial transient under adversarial training. This comparison is descriptive: the purpose is to determine whether InterpScore simply mirrors instantaneous fluctuations in predictive performance or exhibits a distinct representation-level trajectory. The resulting benchmark scores are summarized in Figure 6.

Figure 6 shows 95% confidence intervals for InterpScore across architecture and training-regime pairings, with standard training in blue and adversarial train-

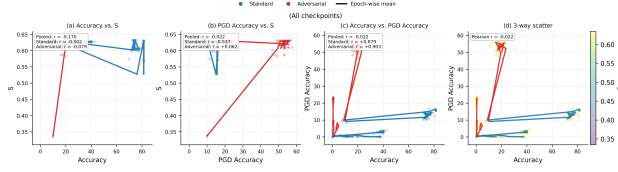


Figure 7: Relationships between InterpScore, predictive accuracy, and adversarial robustness across the evaluated model configurations. The analysis examines whether InterpScore is empirically reducible to conventional task-performance measures.

ing in red. All mean estimates cluster tightly between about 0.627 and 0.633, and the intervals overlap substantially. The ResNet-18 rows at the top have the highest means but also the widest intervals, while one ResNet-18 rows at the top have the highest means but also the widest intervals, while one ResNet-18 adversarial configuration has a very narrow interval, indicating low replicate variance. This is exactly what a leaderboard does: it places every model and training regime on a single common scale with explicit uncertainty, enabling scalable, reproducible, architecture-agnostic comparison.

4.9 RQ4: Complementarity with Predictive Accuracy and Robustness

We finally examine whether InterpScore captures information that is not reducible to predictive accuracy or adversarial robustness. Figure 7 summarizes the empirical relationships among InterpScore, predictive accuracy, and robustness.

Across checkpoints, InterpScore shows no clear monotonic relationship with either standard accuracy or PGD accuracy. Accuracy and PGD accuracy are strongly related to each other, but InterpScore remains comparatively stable across their range. Thus, InterpScore appears to capture information not reflected directly in accuracy or robustness, although the current correlations are descriptive and insufficient to establish independence.

Finally, we examine the joint structure of robustness and interpretability through Pareto analysis (Figure 8). The configurations occupy distinct, non-collapsed positions in the joint robustness-interpretability space. Models with similar (or even identical) PGD robustness can exhibit different InterpScore values, indicating that the two quantities are not necessarily interchangeable. This analysis is descriptive rather than an optimization claim. In particular, we do not use the joint plot to identify a universally preferred operating point. Instead, the purpose is to determine whether In-

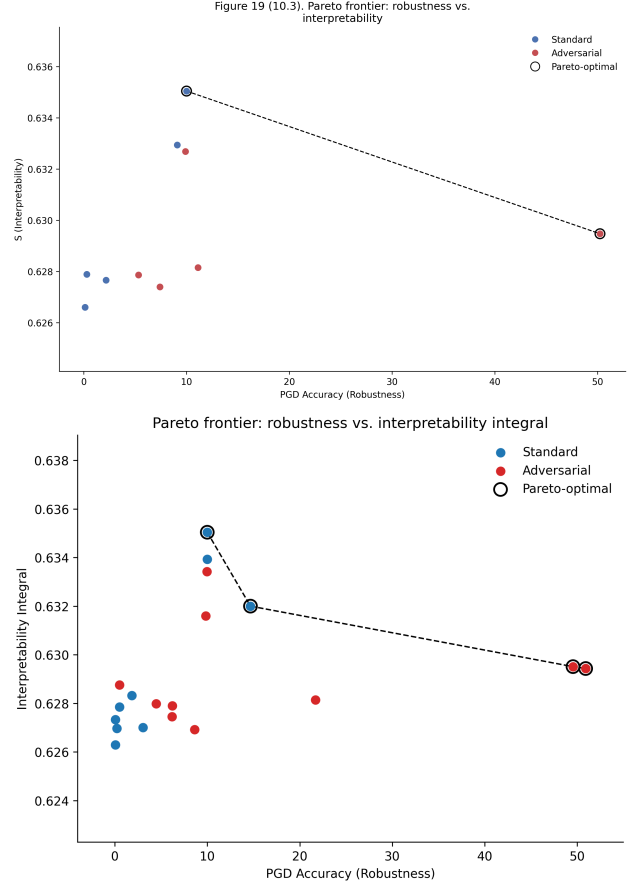


Figure 8: Robustness-interpretability relationship on the large-scale benchmark. Each point corresponds to an evaluated model configuration, with PGD accuracy and InterpScore providing complementary axes for characterizing learned models. The analysis is descriptive and does not define a preferred operating point.

terpScore provides a distinct axis along which learned representations can be characterized.

5 DISCUSSION

Our results provide initial evidence that mechanistic interpretability can be studied through quantitative properties of learned representations. InterpScore is sensitive to controlled changes in representational capacity while remaining only weakly related to conventional accuracy and adversarial robustness, suggesting that it captures a complementary representation-level property rather than simply remeasuring predictive performance. The results further motivate our central hypothesis that *feature superposition* may constitute a latent geometric variable connecting interpretability and adversarial robustness. In this view, successful ad-

versarial perturbations serve as a reproducible probe of representation geometry, while superposition is the hypothesized underlying property influencing how features are organized and how perturbations propagate through the representation.

Several limitations qualify this interpretation. First, InterpScore is not intended to provide a complete measure of mechanistic interpretability, which also involves semantic features, causal mechanisms, and circuit structure. Second, the experiments focus on image classification and therefore do not establish generality across modalities or model scales. Third, the relationship between superposition, interpretability, and robustness remains correlational: our experiments do not establish that reducing superposition causes either improved interpretability or robustness. Finally, the metric depends on design choices concerning adversarial perturbations and geometric descriptors, which should be tested more broadly.

Despite these limitations, the work suggests a path toward more standardized quantitative evaluation of mechanistic interpretability. Rather than treating interpretability solely as a qualitative property requiring model-specific inspection, representation-level quantities can be measured, compared, and validated against controlled ground truth. If future work establishes a causal relationship between feature superposition, representation geometry, and interpretability, such measurements could provide a common quantitative framework for studying interpretability alongside robustness and other properties of learned models.

6 FUTURE DIRECTION

Our results suggest a broader hypothesis connecting feature superposition, adversarial vulnerability, interpretability, and architectural capacity. If superposition is a common geometric factor underlying both adversarial behavior and interpretability, then representation geometry may provide not only a way to measure interpretability, but also a signal for how networks should be trained and when their architectures should grow.

A natural next step is to develop more direct quantitative measures of superposition. Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$ denote the ordered eigenvalues of the representation covariance matrix. A rapidly decaying spectrum indicates that representational variance is concentrated in a small number of directions, whereas a flatter spectrum indicates that variance is distributed across many directions and is consistent with stronger superposition. Effective rank can summarize the number of directions carrying substantial variance, while a normalized PCA score can quantify the fraction cap-

tured by the dominant components. An important question is whether these spectral quantities and the geometric structure of successful adversarial representations provide complementary measurements of the same underlying representational organization.

These measurements could also make it possible to monitor superposition throughout training rather than evaluating interpretability only after convergence. Increasing effective dimensionality, flattening eigenvalue spectra, or increasing geometric complexity could serve as indicators that a representation is becoming increasingly polysemantic. Such a signal could turn interpretability measurement into a mechanism for architectural adaptation. Instead of expanding a network according to a fixed schedule or optimization heuristic, additional neurons, channels, or branches could be introduced where the representation exhibits evidence of a structural bottleneck. Regions whose representations remain sufficiently concentrated would require no additional capacity, while regions exhibiting increasing superposition could receive dedicated representational resources. Architectural growth would therefore be driven by representational structure rather than prescribed independently of it.

This idea could be combined with active learning: acquisition functions could determine *which examples* are most informative, while structural measurements determine *where capacity* is needed. Uncertainty, diversity, representativeness, or expected model change could therefore be coupled with representation-level bottleneck signals to jointly adapt the training trajectory and architecture.

Several questions remain open. Effective rank, PCA concentration, and related geometric quantities are proxies for representational structure rather than complete measures of interpretability: low-dimensional representations need not be semantically meaningful, while high-dimensional representations need not be mechanistically incoherent. Complementary measurements of semantic breadth, feature sparsity, circuit modularity, and causal faithfulness will therefore remain important. The proposed relationships should also be tested across architectures, scales, datasets, and objectives, particularly in transformer and language-model representations, and it remains to be established whether adversarially probed geometry provides information beyond naturally occurring activations.

The longer-term goal is to move from *measuring* superposition to *controlling* it. If increasing superposition can be detected reliably, training could respond by allocating capacity to emerging feature directions before representations become strongly polysemantic.

This could enable networks whose internal structure remains more modular as they grow, providing representations that are subsequently easier to analyze using causal interpretability tools.

Overall, this suggests a progression from measurement to diagnosis to adaptation: quantify superposition, use its evolution to identify structural bottlenecks, and ultimately use those signals to allocate representational capacity. This would move interpretability from a property measured after training toward a structural quantity that can potentially inform how neural networks are trained and grown.

AI use statement

In this work, we used generative AI tools for language editing, proofreading, improving the clarity of the manuscript, and assisting with code development.

We have not used generative AI tools for research design, formulation of the research questions, development of the scientific methodology, generation or interpretation of experimental results, or the formulation of scientific claims, and the rest of the required disclosure tasks are not applicable to this work.

Additionally, we used generative AI tools for improving the wording and readability of the manuscript and for assisting with implementation and debugging of code.

We have reviewed all AI-assisted work. All AI-assisted text was carefully reviewed and edited by the authors. AI-assisted code was inspected, tested, and verified by the authors, including checking that its behavior and outputs were consistent with the intended methodology and experimental setup. The authors independently verified the reported experimental results and conclusions.

We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

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All acknowledgements go at the end of the paper, including thanks to reviewers who gave useful comments, to colleagues who contributed to the ideas, and to funding agencies and corporate sponsors that provided financial support. To preserve the anonymity, please include acknowledgements *only* in the camera-ready papers. The acknowledgements do not count against the 9-page page limit in the camera-ready.

References

References follow the acknowledgements. Use an unnumbered third level heading for the references section. Please use the same font size for references as for the body of the paper—remember that references do not count against your page length total.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. Yes
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. Yes
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. Yes
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. Yes
 - (b) Complete proofs of all theoretical results. Yes
 - (c) Clear explanations of any assumptions. Yes

3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). Yes
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). Yes
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). Yes
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). Yes
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
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 - (b) The license information of the assets, if applicable. Not Applicable
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 - (d) Information about consent from data providers/curators. Not Applicable
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. Not Applicable
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. Not Applicable
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. Not Applicable
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. Not Applicable

From Adversarial Robustness to Interpretability: Toward Quantitative Metrics for Transparent Artificial Intelligence

A THEORY

A.1 Setup and Notation

Let $f_\theta : \mathcal{X} \rightarrow \{1, \dots, C\}$ be a classifier with representation map $\phi_\ell : \mathcal{X} \rightarrow \mathbb{R}^{d_\ell}$ at layer ℓ , where $\mathcal{X} \subset \mathbb{R}^{d_0}$. For $x \in \mathcal{X}$ with true label y and perturbation budget $\epsilon > 0$ under a norm $\|\cdot\|$, define the attack ball

$$B_\epsilon(x) = \{x + \delta : \|\delta\| \leq \epsilon\}$$

and the **success set**

$$\mathcal{A}_\epsilon(x) = \{\delta : \|\delta\| \leq \epsilon, f_\theta(x + \delta) \neq y\}.$$

An attack algorithm (PGD, AutoAttack, ...) with m random restarts returns a finite sample $\hat{\mathcal{A}}_\epsilon(x) = \{\delta_1, \dots, \delta_m\} \subset \mathcal{A}_\epsilon(x)$. If $\hat{\mathcal{A}}_\epsilon(x) = \emptyset$ we set the score below to its maximal value by convention, since an exhausted attack budget with no success is itself evidence of local robustness.

Pushing successful perturbations through the network gives the **adversarial representation manifold**

$$M_\ell(x, \epsilon) = \{\phi_\ell(x + \delta_i)\}_{i=1}^m \subset \mathbb{R}^{d_\ell},$$

aggregated over a dataset sample $\{x_1, \dots, x_N\}$ as

$$M_\ell(\epsilon) = \bigcup_j M_\ell(x_j, \epsilon).$$

[Adversarial Hull]

$$H_\ell(\epsilon) := (M_\ell(\epsilon)) \subset \mathbb{R}^{d_\ell}.$$

Literal high-dimensional polytope volume is numerically unstable and not scale-invariant, so we characterize H_ℓ through the second-moment structure of its generating point cloud rather than its volume directly.

[Interpretability Score] Let

$$\Sigma_\ell(\epsilon) = (M_\ell(\epsilon))$$

and $m_\ell = |M_\ell(\epsilon)|$. Define

$$I_\ell(\epsilon) = 1 - \frac{(\Sigma_\ell(\epsilon))}{\min(m_\ell, d_\ell)} \in [0, 1].$$

A.2 Basic Properties

[Well-Posedness] $I_\ell(\epsilon) \in [0, 1]$. Moreover $I_\ell(\epsilon) = 1$ iff $M_\ell(\epsilon)$ lies on a single line through its centroid (a rank-1 hull — every successful attack displaces the representation along one direction), and $I_\ell(\epsilon) \rightarrow 0$ as $\Sigma_\ell(\epsilon) \rightarrow$ isotropic (successful attacks equally likely along every direction).

$H(p)$ is maximized at the uniform distribution ($\log \min(m_\ell, d_\ell)$) and minimized at a point mass ($H = 0$); exponentiating gives $\in [1, \min(m_\ell, d_\ell)]$, and the stated bound on I_ℓ follows immediately.

[Orthogonal Invariance] For any $U \in O(d_\ell)$, computing I_ℓ from $U \cdot M_\ell(\epsilon)$ leaves $I_\ell(\epsilon)$ unchanged.

Covariance eigenvalues are invariant under orthogonal change of basis; depends on Σ only through its eigenvalues.

This is not a cosmetic property. Any metric claimed to measure representational “interpretability” as a geometric fact about a layer — rather than an artifact of how individual neurons happen to be indexed — must be invariant to rotation of the representation basis. I_ℓ satisfies this; naive per-neuron sparsity statistics generally do not.

[Monotonicity in budget — a caveat, not a theorem] One might expect $I_\ell(\epsilon)$ to be monotone in ϵ under nested success sets $\mathcal{A}_{\epsilon_1} \subseteq \mathcal{A}_{\epsilon_2}$. This does *not* follow automatically: a larger budget can either recruit genuinely new attack directions (lowering I_ℓ) or simply scale existing ones (leaving unchanged). We flag this as an open empirical question rather than a proven monotonicity result — the depth/width plots in the talk should be read as evidence about *typical* behavior, not as a consequence of Definition 1.3.

A.3 The Central Bridge: From Hull Geometry to Superposition

[Hull–Superposition Correspondence] Suppose the layer- ℓ representation admits a superposition decomposition

$$\phi_\ell(x) = \sum_{k=1}^K a_k(x) v_k$$

over a (possibly overcomplete) feature dictionary $\{v_k\} \subset \mathbb{R}^{d_\ell}$ (Elhage et al., 2022), and suppose the successful-attack sample $M_\ell(\epsilon)$ activates exactly $r \leq K$ dictionary directions with non-vanishing coefficient variance while the remaining $K - r$ directions have coefficient variance $\rightarrow 0$ — i.e., attacks only ever displace the representation along r “vulnerable” feature directions. Then

$$(\Sigma_\ell(\epsilon)) \rightarrow r \implies I_\ell(\epsilon) \rightarrow 1 - \frac{r}{\min(m_\ell, d_\ell)}.$$

As the coefficient variance on the $K - r$ inactive directions vanishes, the corresponding eigenvalues of $\Sigma_\ell(\epsilon) \rightarrow 0$, so $p_i \rightarrow 0$ for those indices and their contribution to $H(p)$ vanishes; the entropy converges to that of the r -dimensional active subspace’s spectrum, whose is at most r and equals r in the isotropic-within-subspace case.

This theorem makes precise, under an explicit and checkable modeling assumption, the informal claim that “adversarial examples concentrate . . . where superposition is highest”: the number of feature directions r that attacks are able to exploit is exactly what I_ℓ measures. It is a *conditional* result — it explains why I_ℓ should track superposition given the toy-model decomposition, not a proof that real networks satisfy that decomposition.

A.4 A Robustness Bridge (Adapted, Not Novel)

[Volume–Certified-Radius Correspondence] Let

$$g(x) = \arg \max_c \Pr_{\xi \sim \mathcal{N}(0, \sigma^2 I)} [f_\theta(x + \xi) = c]$$

be the randomized-smoothed classifier (Cohen, Rosenfeld & Kolter, 2019). If the fraction of Gaussian noise mass landing in the success region around x is bounded, $p \leq \bar{p}$, then g is certifiably correct at x up to radius

$$R = \sigma \Phi^{-1}(1 - \bar{p}),$$

where Φ is the standard normal CDF. Since a smaller, thinner adversarial hull $H_\ell(\epsilon)$ (equivalently, a larger $I_\ell(\epsilon)$) corresponds to a smaller pre-image success region in input space, networks with uniformly high $I_\ell(\epsilon)$ over their data support are expected to admit larger certified radii under smoothing.

This is an adaptation of an established certification result, not a new theorem — the correspondence between hull thinness and $\bar{p}$ is heuristic (they are related but not identical objects: one lives in representation space, the other in input space). We include it because it is the cleanest existing bridge between “geometric fingerprint of vulnerability” and a certifiable robustness guarantee, and it suggests a concrete follow-up experiment: regress empirical certified radius against I_ℓ across the architecture sweep already run for the CIFAR-10/SVHN figures.

[Limitation] I_ℓ is a second-moment surrogate for hull geometry. It cannot detect a success region that is genuinely multi-modal (e.g., two disjoint thin clusters in orthogonal directions) but has zero net correlation — such a region would register the same as a single isotropic-in-2D cluster. This is the same blind spot every covariance-based metric inherits and is worth stating explicitly rather than leaving implicit.

Table 2: Principal notation used in the proposed benchmark.

Symbol	Definition
$\mathcal{D}$	evaluation dataset
N	number of evaluated examples
C	number of output classes
L	number of analyzed layers
d_l	original representation dimensionality at layer l
k	PCA dimensionality
f	trained classifier
ϕ_l	representation mapping at layer l
x_n	original input
x_n^{adv}	adversarially perturbed input
c_t	true class of the input
c_p	adversarially predicted class
$z_n^{(l)}$	reduced layer- l representation
$r_n^{(l)}$	PCA representation
$\mathcal{G}_{c_t, c_p}^{(l)}$	transition-specific reservoir
$\mathcal{H}_{c_t, c_p}^{(l)}$	transition-specific convex hull
G_l	within-manifold geometry descriptor
O_l	mean pairwise overlap
S_l	mean pairwise separation
$\mathcal{B}_l$	layer-wise burden
$\mathcal{I}_l$	layer-wise interpretability score
$\mathcal{I}_{\text{int}}$	integrated network-level score
α	interaction weight
Q	separation amplification parameter
M	reservoir capacity
T	number of PGD iterations
m	Monte Carlo samples per hull comparison

B METHODOLOGICAL PROPERTIES

B.1 Notation

Table 2 summarizes the principal notation used throughout the method.

B.1.1 Inputs and Outputs

The benchmark takes as input:

1. a trained classifier f ;
2. an evaluation dataset $\mathcal{D}$;
3. a set of representation layers $\{1, \dots, L\}$;
4. an adversarial attack configuration;
5. a dimensionality-reduction configuration; and
6. geometric and score hyperparameters.

The primary outputs are:

1. a layer-wise interpretability profile $\{\mathcal{I}_1, \dots, \mathcal{I}_L\}$;

2. the integrated interpretability measure $\mathcal{I}_{\text{int}}$;
3. the underlying geometric descriptors (G_l, O_l, S_l) ; and
4. the Target Transition Matrix $T^{(l)}$.

B.1.2 Objective

The methodological objective is to obtain a quantitative, representation-level measurement whose value depends on the geometry of learned representations associated with successful adversarial transitions.

Specifically, the method seeks to measure whether independently observable changes in representation geometry correspond to independently validated changes in interpretability, while avoiding dependence on human semantic annotations or externally trained probes.

The method therefore does not define interpretability through a particular semantic ontology. Instead, it operationalizes interpretability through measurable geometric properties of the learned representation space.

B.2 Assumptions

The benchmark makes explicit assumptions at four levels: data, distributional, architectural, and computational.

B.2.1 Data Assumptions

The evaluation dataset consists of labeled examples

$$(x_n, y_n) \sim \mathcal{D}. \quad (12)$$

The classifier is assumed to correctly classify the original example before adversarial probing:

$$f(x_n) = y_n. \quad (13)$$

Only examples for which the attack produces a successful change in prediction are used to construct transition manifolds:

$$f(x_n^{\text{adv}}) \neq y_n. \quad (14)$$

Consequently, the measured geometry is conditional on the successful attack distribution rather than on the complete evaluation dataset.

B.2.2 Distributional Assumptions

No parametric probability distribution is imposed on the learned representations.

The empirical transition reservoirs

$$\mathcal{G}_{c_t, c_p}^{(l)} \quad (15)$$

are treated as finite samples from the corresponding conditional representation distributions induced by the data and adversarial procedure.

The method therefore assumes that the retained successful adversarial examples provide sufficient coverage of the transition regions for the geometric quantities to be estimated reliably.

B.2.3 Architectural Assumptions

The classifier must expose a finite collection of intermediate representations

$$\phi_1, \dots, \phi_L \quad (16)$$

that can be extracted during a forward pass.

For convolutional representations, the method assumes that spatial aggregation provides a meaningful vector representation:

$$z_n^{(l)} = \frac{1}{H_l W_l} \sum_{h=1}^{H_l} \sum_{w=1}^{W_l} a_{n,:,h,w}^{(l)}. \quad (17)$$

For transformer representations, the current implementation uses the CLS representation:

$$z_n^{(l)} = a_{n,0,:}^{(l)}. \quad (18)$$

No additional normalization or clipping is applied to the CLS vector.

These reductions are representation-extraction choices rather than claims that spatial averaging or CLS pooling is universally optimal.

B.2.4 Computational Assumptions

The method assumes that the original representation dimensionalities d_l can be reduced to a common dimension k before geometric analysis.

The benchmark also assumes that the transition reservoirs can be bounded by a fixed capacity M :

$$\left| \mathcal{G}_{c_t, c_p}^{(l)} \right| \leq M. \quad (19)$$

This converts the memory requirement of the geometric stage from one that grows with the complete dataset to one bounded by the number of layers, transitions, and reservoir capacity.

Finally, pairwise geometric estimation is assumed to be computationally feasible for the number of observed transition manifolds. A light evaluation mode is provided when this assumption does not hold.

B.3 Ablations and Computational Regimes

B.3.1 Additive Formulation

To distinguish interaction effects from simple aggregation, we consider the additive alternative

$$\boxed{\mathcal{B}_l^{\text{add}} = G_l + O_l (1 + Q S_l^{\text{norm}})}. \quad (20)$$

The corresponding interactive formulation is

$$\boxed{\mathcal{B}_l^{\text{int}} = G_l [1 + O_l (1 + Q S_l^{\text{norm}})]}. \quad (21)$$

The comparison tests whether overlap and separation contribute independently of intrinsic geometric complexity or whether their effect should scale with the geometry of the transition regions.

This additive formulation is therefore treated as an ablation rather than as an alternative primary definition.

B.3.2 Light Evaluation Mode

For large-scale benchmarking, a computationally lighter score can be obtained by evaluating only the within-manifold geometry:

$$\boxed{\mathcal{I}_l^{\text{light}} = 1 - G_l}. \quad (22)$$

This mode avoids the pairwise overlap and separation computation and therefore removes the quadratic dependence on the number of transition manifolds.

The full and light scores are consequently treated as distinct computational regimes rather than mathematically identical transformations.

B.4 Network-Level Aggregation

The layer-wise score produces a representation-depth profile

$$\{\mathcal{I}_1, \dots, \mathcal{I}_L\}. \quad (23)$$

To account for representation depth, layer positions are normalized as

$$t_l = \frac{l-1}{L-1}, \quad t_l \in [0, 1]. \quad (24)$$

The integrated interpretability measure is defined as

$$\boxed{\mathcal{I}_{\text{int}} = \int_0^1 \mathcal{I}(t) dt.} \quad (25)$$

It is numerically approximated using the trapezoidal rule:

$$\boxed{\mathcal{I}_{\text{int}} \approx \sum_{l=1}^{L-1} \frac{\mathcal{I}_l + \mathcal{I}_{l+1}}{2} (t_{l+1} - t_l).} \quad (26)$$

Since

$$t_{l+1} - t_l = \frac{1}{L-1}, \quad (27)$$

this becomes

$$\mathcal{I}_{\text{int}} = \frac{1}{L-1} \sum_{l=1}^{L-1} \frac{\mathcal{I}_l + \mathcal{I}_{l+1}}{2}. \quad (28)$$

Design choice. Depth integration is a numerical aggregation procedure rather than a new geometric primitive. It is used to preserve information about the full representation trajectory rather than reducing the network to a single arbitrarily selected layer.

B.5 Target Transition Matrix

In addition to the adversarial transition manifolds, we compute a complementary class-conditioned geometric descriptor.

For class i , define the class-conditioned representation hull at layer l as

$$\mathcal{Q}_i^{(l)} = \text{Conv} \left(\left\{ r_n^{(l)} : y_n = i \right\} \right). \quad (29)$$

Let its scalar geometric descriptor be

$$g_i^{(l)} = 0.40 \bar{V}_i^{(l)} + 0.30 B_i^{(l)} + 0.30 \rho_i^{(l)}, \quad (30)$$

where $\rho_i^{(l)}$ denotes the corresponding normalized vertex-density descriptor.

The Target Transition Matrix (TTM) is defined as

$$\boxed{T_{ij}^{(l)} = g_j^{(l)} - g_i^{(l)}.} \quad (31)$$

This quantity is a directed geometric contrast rather than a metric distance. Consequently,

$$T_{ji}^{(l)} = g_i^{(l)} - g_j^{(l)} = -T_{ij}^{(l)}, \quad (32)$$

and

$$T_{ii}^{(l)} = 0. \quad (33)$$

The matrix is therefore antisymmetric:

$$T^{(l)} = -\left(T^{(l)}\right)^\top. \quad (34)$$

A multidimensional alternative retains the complete geometric descriptor vector:

$$\mathbf{g}_i^{(l)} = \begin{bmatrix} \bar{V}_i^{(l)} \\ B_i^{(l)} \\ \rho_i^{(l)} \end{bmatrix}. \quad (35)$$

The corresponding directed geometric contrast is

$$\Delta \mathbf{g}_{ij}^{(l)} = \mathbf{g}_j^{(l)} - \mathbf{g}_i^{(l)}. \quad (36)$$

The TTM is complementary to the primary score. The primary score summarizes the geometry of successful adversarial transition manifolds, whereas the TTM summarizes class-conditioned geometric contrasts.

B.6 Computational Complexity and Cost

Let N denote the number of evaluated examples, L the number of analyzed layers, k the PCA dimensionality, M the maximum reservoir size, R_l the number of valid transition manifolds at layer l , m the number of Monte Carlo samples used for overlap estimation, and T the number of PGD iterations.

The dominant cost of adversarial generation is approximately

$$O(NTC_f), \quad (37)$$

where C_f denotes the cost of a forward/backward model evaluation.

Streaming PCA avoids storing the complete activation matrix $N \times d_l$. Its exact computational cost depends on the streaming PCA implementation and the dimensionality of the incoming representations.

The bounded transition reservoirs require at most

$$O(LR_l Mk) \quad (38)$$

storage for layer- l transition representations, summed over layers and valid transitions.

At layer l , the number of pairwise hull comparisons is

$$P_l = \binom{R_l}{2} = \frac{R_l(R_l - 1)}{2}, \quad (39)$$

which yields

$$O(R_l^2) \quad (40)$$

pairwise comparisons.

Monte Carlo overlap estimation introduces an additional factor m :

$$O(R_l^2 m). \quad (41)$$

Consequently, the pairwise geometric stage becomes the dominant component when R_l is large.

The bounded reservoirs ensure that memory consumption during geometric analysis does not grow with the total number of processed examples once reservoir capacity has been reached.

Computational interpretation. The benchmark therefore has two distinct computational bottlenecks: adversarial generation scales with the number of evaluated examples and attack iterations, whereas pairwise geometric estimation scales quadratically with the number of observed transition manifolds.

B.7 Design Choices and Justification

The method combines established computational primitives with components introduced specifically for the proposed measurement framework.

B.7.1 Successful Adversarial Transitions

Choice. Representations are conditioned on successful transitions

$$c_t \rightarrow c_p \quad (42)$$

rather than pooled across all adversarial examples.

Justification. This preserves the direction and identity of the observed failure mode. A global adversarial representation distribution would combine different target classes and therefore discard information about which decision boundary was crossed.

Status. New methodological component.

B.7.2 Transition-Specific Reservoirs

Choice. Representations are stored separately for every ordered transition (c_t, c_p) .

Justification. This preserves class-conditional structure while providing bounded memory through

$$|\mathcal{G}_{c_t, c_p}^{(l)}| \leq M. \quad (43)$$

Status. New methodological component.

B.7.3 Low-Dimensional Geometric Representation

Choice. Geometric analysis is performed after projection into a common k -dimensional PCA space.

Justification. Direct geometric computation in the original representation spaces can be computationally prohibitive and can make comparisons across layers difficult because d_l varies with depth.

Status. PCA is **borrowed**; its use as the common coordinate system for transition-manifold geometry is **adapted**.

B.7.4 Convex Hull Approximation

Choice. Each finite transition reservoir is represented by its convex hull.

Justification. The convex hull provides an explicit finite geometric region from which volume, boundary, vertex, intersection, and separation quantities can be estimated.

Alternative representations such as kernel density estimates or alpha shapes could preserve non-convex structure more explicitly, but would introduce additional smoothing or shape parameters.

Status. Convex-hull construction is **borrowed**; its application to adversarial transition reservoirs is **adapted**.

B.7.5 Within-Manifold Geometry

Choice. The intrinsic geometry descriptor combines normalized volume, boundary complexity, and vertex density:

$$G_l = 0.40 \bar{V}_l + 0.30 B_l^{\text{norm}} + 0.30 D_l. \quad (44)$$

Justification. A single geometric statistic cannot characterize all aspects of a finite representation region. The three descriptors capture complementary aspects of spatial extent and boundary structure.

Status. The individual geometric quantities are **borrowed or adapted**; their combination into G_l is **new**.

B.7.6 Overlap and Separation

Choice. Pairwise overlap and Euclidean separation are computed between transition manifolds.

Justification. Within-manifold geometry does not determine how different regions are organized relative to one another. Overlap and separation provide complementary measures of their pairwise organization.

Status. The geometric primitives are **borrowed or adapted**; their integration with transition-specific adversarial manifolds is **new**.

B.7.7 Interactive Score

Choice. The primary burden is

$$\mathcal{B}_l = G_l [1 + \alpha O_l (1 + Q S_l^{\text{norm}})]. \quad (45)$$

Justification. The interaction formulation tests whether between-manifold effects should scale with intrinsic geometric complexity rather than contribute as an independent additive quantity.

The additive formulation in Eq. (20) is therefore retained as an ablation.

Status. **New.**

B.7.8 Depth Integration

Choice. The layer-wise score is integrated over normalized representation depth.

Justification. A single layer can provide an incomplete description of the learned representation. Depth integration preserves the trajectory of the measurement through the network.

Status. The trapezoidal rule is **borrowed**; its application to the layer-wise interpretability profile is **adapted**.

B.8 Theory and Properties

The following properties follow directly from the definitions under the stated conditions.

B.8.1 Boundedness of the Interpretability Score

Assume

$$G_l \geq 0, \quad O_l \geq 0, \quad S_l^{\text{norm}} \geq 0, \quad \alpha \in [0, 1], \quad Q \geq 0. \quad (46)$$

Then

$$\mathcal{B}_l \geq 0. \quad (47)$$

Therefore,

$$0 < \frac{1}{1 + \mathcal{B}_l} \leq 1, \quad (48)$$

and hence

$$\boxed{\mathcal{I}_l \in (0, 1]}. \quad (49)$$

A proof can be provided in Appendix ??.

Table 3: Principal benchmark hyperparameters and their methodological roles.

Parameter	Default	Role
ϵ	8/255	PGD perturbation budget
α_{PGD}	2/255	PGD step size
T	10	number of PGD iterations
k	10	common PCA dimensionality
M	200	maximum points per transition reservoir
α	0.5	interaction weight
Q	experimental	separation amplification
Standardization	Enabled	coordinate preprocessing before PCA
Successful-only	Enabled	retain only successful adversarial transitions

B.8.2 Monotonicity with Respect to Geometry

For fixed O_l , S_l^{norm} , α , and Q , the burden is linear in G_l :

$$\frac{\partial \mathcal{B}_l}{\partial G_l} = 1 + \alpha O_l (1 + Q S_l^{\text{norm}}) \geq 1. \quad (50)$$

Since

$$\mathcal{I}_l = (1 + \mathcal{B}_l)^{-1}, \quad (51)$$

we obtain

$$\frac{\partial \mathcal{I}_l}{\partial G_l} = -\frac{\partial \mathcal{B}_l / \partial G_l}{(1 + \mathcal{B}_l)^2} \leq 0. \quad (52)$$

Thus, under the non-negativity conditions above, increasing within-manifold geometric burden cannot increase the score.

B.8.3 Antisymmetry of the Target Transition Matrix

From

$$T_{ij}^{(l)} = g_j^{(l)} - g_i^{(l)}, \quad (53)$$

we have

$$T_{ji}^{(l)} = g_i^{(l)} - g_j^{(l)} = -T_{ij}^{(l)}. \quad (54)$$

Therefore,

$$\boxed{T^{(l)} = -\left(T^{(l)}\right)^\top.} \quad (55)$$

This property holds exactly whenever the TTM is defined as the difference of the same scalar class descriptors.

B.9 Complete Benchmark Algorithm.

The complete procedure for one trained model consists of two passes, followed by geometric estimation and score aggregation.

B.10 Hyperparameters and Their Roles

The principal benchmark parameters are summarized in Table 3.

The parameters play different methodological roles.

The attack parameters $(\epsilon, \alpha_{\text{PGD}}, T)$ determine the perturbation distribution used to probe the classifier.

The PCA dimensionality k determines the common geometric space in which the transition manifolds are analyzed.

The reservoir size M controls the memory–sampling trade-off.

The score parameters α and Q determine how the geometric components are combined. In particular,

$$\alpha = 0 \tag{56}$$

removes the interaction contribution, while larger values increase the influence of overlap–separation structure.

The separation amplification parameter Q determines the sensitivity of the overlap–separation term to normalized separation.

Its numerical value should be selected and justified through the corresponding ablation rather than being identified with the number of pairwise hull comparisons.

B.11 Methodological Interpretation

The proposed method does not assume that a single geometric property constitutes interpretability. Instead, it treats interpretability as an empirically testable property of the organization of learned representations.

The within-manifold term G_l captures the intrinsic geometric complexity of transition regions, while O_l and S_l characterize relationships between distinct transition regions.

The resulting layer-wise measurement therefore combines local geometric structure with the organization of multiple representation regions.

The benchmark is designed to answer the following empirical question: Does the measured representation geometry track independently validated changes in interpretability?

The synthetic experiments address this question under controlled ground-truth factors, while the real-world experiments test whether the resulting measurement remains stable and informative across architectures, datasets, and training regimes.

The method consequently separates the *measurement construction* from the *empirical validation* of the measurement. The former is specified by the definitions and algorithms above; the latter is evaluated experimentally through controlled synthetic benchmarks and real-world model comparisons.

B.12 Summary of Methodological Contributions

The methodological contribution can be decomposed into the following components:

1. **Adversarial transition probing:** successful adversarial examples are used to identify directed representation transitions $c_t \rightarrow c_p$.
2. **Transition-specific representation manifolds:** representations are conditioned on ordered class transitions rather than pooled into a global geometric object.
3. **Multi-component geometric characterization:** transition manifolds are described through volume, boundary complexity, and vertex density.
4. **Between-manifold characterization:** pairwise overlap and separation quantify the organization of distinct transition regions.
5. **Interactive interpretability measurement:** within-manifold geometry and between-manifold relationships are combined through the proposed burden formulation.
6. **Depth-wise aggregation:** the layer-wise measurement is integrated across normalized network depth.
7. **Target Transition Matrix:** class-conditioned geometric contrasts provide a complementary directed descriptor of representation organization.

The underlying computational primitives—PGD, forward-hook activation extraction, streaming PCA, convex-hull construction, Monte Carlo estimation, and trapezoidal integration—are established techniques. The contribution of the benchmark lies in their organization into a single quantitative framework for measuring representation geometry through adversarial transitions.

B.13 Within-Manifold Geometry

Each transition manifold is characterized by three normalized geometric descriptors.

B.13.1 Normalized Volume

Let

$$V\left(\mathcal{H}_{c_t, c_p}^{(l)}\right) \quad (57)$$

denote the volume of the transition hull in the k -dimensional PCA space.

The volume is normalized relative to the reference representation region used by the benchmark:

$$\bar{V}_{c_t, c_p}^{(l)} \in [0, 1]. \quad (58)$$

The layer-wise volume descriptor is

$$\bar{V}_l = \frac{1}{R_l} \sum_{q=1}^{R_l} \bar{V}_q^{(l)}, \quad (59)$$

where R_l denotes the number of valid transition manifolds at layer l .

B.13.2 Statistical Reporting

Unless otherwise stated, individual runs are retained rather than reporting only aggregate values. For each experimental condition, we report the mean together with an appropriate measure of variability across independent runs. Where confidence intervals are reported, the procedure used to construct the interval is specified explicitly.

For comparisons involving repeated measurements of the same trained models, paired analyses are used where appropriate. For comparisons involving independent experimental units, independent-sample analyses are used instead. When multiple hypotheses are tested simultaneously, the corresponding multiplicity correction is reported.

In addition to statistical significance, we report effect sizes where the number of independent experimental units permits meaningful estimation. This distinction is important because a statistically detectable difference does not by itself establish practical importance.

B.13.3 Implementation and Computational Budget

All models are trained using the same dataset construction and preprocessing protocol within each experimental condition. For the α ablation, the trained networks and extracted representations are held fixed; no model is retrained for different values of α .

The synthetic benchmark therefore uses the same training budget across architectures and bottleneck dimensionalities, with the random seed being the primary source of stochastic variation between replicated runs. The large-scale comparison similarly applies the same training and evaluation protocol within each architecture–dataset–training-regime configuration.

All hyperparameters used to compute InterpScore are fixed before evaluation of the corresponding test results. In particular, α is not selected by optimizing agreement with the synthetic ground-truth characterization.

B.14 Reproducibility

All experiments are designed to permit reproduction from the released implementation, dataset-generation procedure, configuration files, and random seeds.

For the synthetic benchmark, the dataset construction procedure and generative factors are fixed and recorded explicitly. The five random seeds used for each architecture and bottleneck dimensionality are retained so that the 50 independently trained models can be regenerated.

For the large-scale benchmark, the architecture, dataset, training regime, evaluation configuration, and random seeds are specified for each experimental condition. The same evaluation pipeline is applied across standard and adversarially trained models.

The released configuration specifies the InterpScore hyperparameters, including the descriptor weighting parameter α , PCA dimensionality, transition-reservoir size, overlap-estimation budget, and adversarial-evaluation parameters where applicable. No test-set result is used to tune these parameters.

The implementation additionally records the software environment and computational configuration required to reproduce the experiments, including the use of an NVIDIA A100 GPU, 16 CPUs, and 128 GB of RAM. Reported aggregate values are accompanied by the underlying per-run measurements wherever possible, allowing the reported uncertainty, statistical tests, and effect sizes to be independently recomputed.

C EXPERIMENTAL CONFIGURATIONS AND ADDITIONAL QUANTITATIVE RESULTS

This appendix provides the experimental configurations and complete numerical results underlying the experiments reported in the main paper. The tables are intended to complement the main-paper figures by providing the exact experimental settings, per-configuration quantitative results, and the sensitivity of InterpScore to its weighting parameter. Unless otherwise stated, all reported values are computed using the same evaluation protocol described in the main paper.

C.1 Experimental Configurations

Table 4 summarizes the experimental configurations used throughout the benchmark. We report the architecture, dataset, training regime, optimizer, learning rate, batch size, number of training epochs, weight decay, and the parameters used for adversarial training where applicable. For each configuration, two independent training runs are performed in the large-scale benchmark.

Table 4: Experimental configurations used in the large-scale benchmark. Standard training uses the default data augmentation and optimization procedure for the corresponding architecture and dataset. Adversarial training uses projected gradient descent (PGD). Values are reported for reproducibility; entries marked with “—” are not applicable.

Architecture	Dataset	Training	Runs
ResNet-18	CIFAR-10	Standard	2
ResNet-18	CIFAR-10	PGD adversarial	2
ResNet-18	CIFAR-100	Standard	2
ResNet-18	CIFAR-100	PGD adversarial	2
ResNet-18	TinyImageNet-200	Standard	2
ResNet-18	TinyImageNet-200	PGD adversarial	2
VGG-16	CIFAR-10	Standard	2
VGG-16	CIFAR-10	PGD adversarial	2
ViT-B/16	CIFAR-100	Standard	2
ViT-B/16	CIFAR-100	PGD adversarial	2

For completeness, the optimization and adversarial-training hyperparameters are given in Table 5. These values should be populated directly from the training configuration used to produce the reported checkpoints.

C.2 Complete Quantitative Benchmark Results

Table 6 reports the complete numerical results for the large-scale benchmark. We include both task performance and interpretability measures so that the relationship between predictive performance, adversarial robustness, and representation interpretability can be examined directly. Accuracy and PGD accuracy measure standard

Table 5: Optimization and evaluation hyperparameters used in the experiments. “—” denotes a parameter that is not applicable to the corresponding training regime.

Hyperparameter	Value
Optimizer	Stochastic Gradient Descent
Momentum	0.9
Learning rate	0.001
Batch size	32/128
Number of epochs	100
Weight decay	0.0005 (L2 regularization coefficient)
PGD perturbation budget ϵ	8/255
PGD step size	2/225
PGD steps	10
InterpScore α	0.75
Number of evaluation runs	2

Architecture	Dataset	Training	Run	Accuracy	PGD Acc.	InterpScore	InterpScore-Integral
ResNet-18	CIFAR-10	Standard	AVE	78.21 ± 29.61	14.52 ± 27.51	0.61 ± 0.03	0.61 ± 0.08
ResNet-18	CIFAR-10	PGD	AVE	20.43 ± 1.78	50.53 ± 51.40	0.62 ± 0.18	0.61 ± 0.25
ResNet-18	CUB-200	Standard	AVE	29.70 ± 105.04	0.08 ± 0.33	0.63 ± 0.00	0.63 ± 0.00
ResNet-18	CUB-200	PGD	AVE	0.68 ± 1.86	9.08 ± 5.70	$0.63 \pm \text{N/A}$	$0.63 \pm \text{N/A}$
ResNet-18	TinyImageNet-200	Standard	AVE	35.82 ± 39.07	2.45 ± 8.20	0.63 ± 0.00	0.63 ± 0.00
ResNet-18	TinyImageNet-200	PGD	AVE	0.66 ± 0.06	21.04 ± 5.78	0.63 ± 0.00	0.63 ± 0.00
VGG-16	CIFAR-10	Standard	AVE	81.71 ± 2.10	15.64 ± 1.65	0.60 ± 0.39	0.60 ± 0.39
VGG-16	CIFAR-10	PGD	AVE	17.30 ± 17.28	51.10 ± 35.64	0.61 ± 0.26	0.61 ± 0.26
ViT-B/16	CUB-200	Standard	AVE	15.77 ± 52.96	0.15 ± 0.55	0.63 ± 0.00	0.63 ± 0.00
ViT-B/16	CUB-200	PGD	AVE	3.07 ± 11.18	7.51 ± 13.82	N/A	N/A

Table 6: Complete quantitative results for the large-scale benchmark. Accuracy, PGD accuracy, InterpScore, and InterpScore-Integral are reported as mean $\pm$ standard deviation over two independent training runs. PGD accuracy denotes accuracy under the adversarial attack used for the corresponding robustness evaluation. Values correspond to the results reported in Table 1 and are rounded to two decimal places.

and adversarial predictive performance, respectively, while InterpScore and InterpScore-Integral quantify the corresponding representation properties.

All quantities are reported as mean $\pm$ standard deviation across the two independent runs for each configuration. The table therefore preserves the run-to-run variability that is otherwise summarized graphically in the main paper.

For the main comparisons, the two independent runs are aggregated within each architecture–dataset–training configuration. This aggregation avoids treating repeated measurements of the same experimental configuration as independent model families while retaining the observed run-to-run variability.

C.3 Training Dynamics and Stability

We first provide the complete training trajectories for the large-scale benchmark. Figure 9 reports accuracy, adversarial accuracy, InterpScore, and InterpScore-Integral throughout training for all evaluated model–dataset configurations.

To further examine the dependence of the measurements on the training stage, Figure 10 compares scores obtained at the best and final checkpoints. This comparison tests whether the conclusions are sensitive to the particular checkpoint used for evaluation.

C.4 Score Distributions and Run-to-Run Variability

The main text reports aggregate scores across the evaluated configurations. Here, Figure 11 shows the corresponding score distributions, exposing the variation across individual runs and model configurations. These

distributions provide additional context for the aggregate comparisons and indicate the degree of variability underlying the reported means and confidence intervals.

We additionally report the complete model-level leaderboard in Figure 6. This provides the individual results underlying the compact comparison presented in the main paper.

C.5 Layer-wise Analysis

Because InterpScore can be evaluated at different stages of a network, we next examine how the measured representation properties vary with network depth. Figure 12 reports layer-wise InterpScore for the evaluated architectures and training regimes.

The corresponding layer-wise training trajectories are shown in Figure 13. These results provide additional information about how the evolution of interpretability depends jointly on training and network depth.

For completeness, Figure 14 reports the individual layer measurements rather than only the aggregate layer-wise summary.

C.6 Relationship to Accuracy and Robustness

We next provide the complete analysis of the relationship between InterpScore, predictive performance, and adversarial robustness. Figure 7 reports the corresponding correlations across the full set of evaluated configurations.

C.7 Rank Stability

Finally, we examine whether comparisons between models remain stable under different evaluation choices. Figure 15 reports rank correlations across the evaluated configurations and scoring variants.

Overall, these additional results provide a detailed characterization of InterpScore beyond the aggregate analyses presented in the main paper. The training trajectories and checkpoint comparisons assess temporal stability, the distribution and layer-wise analyses characterize variability across models and depth, and the correlation, trade-off, Pareto, and rank-stability analyses examine how InterpScore relates to conventional measures of model performance and robustness. Together, these results provide the detailed experimental evidence underlying the more compact presentation in the main paper.

D INTERPBENCH: OPEN-SOURCE IMPLEMENTATION

To facilitate reproducibility and reuse, we provide an open-source implementation of InterpScore through INTERPBENCH. The library contains the implementation of the proposed interpretability metric together with the utilities required to evaluate neural network representations, reproduce the experiments in this paper, and generate the corresponding quantitative analyses and visualizations. The complete implementation and associated experimental resources are publicly available at

D.1 Library Overview

INTERPBENCH is designed as an end-to-end evaluation framework for representation interpretability. Its primary purpose is to provide a consistent interface for computing InterpScore from learned neural representations while separating model training, representation extraction, metric evaluation, and result aggregation. This separation allows the metric to be applied to independently trained models and architectures without requiring modifications to the underlying training procedure.

The repository includes the implementation of InterpScore, experiment scripts, model-training utilities, evaluation and reporting code, and visualization scripts used to produce the results presented in this paper. The repository also contains example scripts and supplementary benchmark resources to facilitate independent use of the framework. The released implementation therefore serves both as the reference implementation for the experiments in this work and as a starting point for evaluating InterpScore on new models and datasets.

D.2 Evaluation Pipeline

The evaluation workflow implemented in INTERPBENCH consists of four main stages. First, a user provides a trained neural network and specifies the representation layer to be evaluated. Second, the corresponding representations are collected over an evaluation dataset. Third, InterpScore is computed from the resulting representation using the geometric and overlap/separation descriptors described in Section 3. Finally, the library aggregates the resulting scores across models, training runs, or experimental conditions and provides utilities for generating tables and figures.

This modular design makes it possible to evaluate multiple architectures using the same metric implementation. In particular, the metric computation does not depend on whether the representation originates from a convolutional network, transformer, or another neural architecture, provided that the corresponding activations can be exposed to the evaluation interface.

D.3 Reproducibility and Experimental Support

The experiments reported in this paper are implemented using the same evaluation code released in INTERPBENCH. The repository contains the scripts used to run experiments, process evaluation outputs, and generate the quantitative reports and figures. This includes the experimental infrastructure used for the synthetic bottleneck experiments and the larger-scale evaluations presented in the paper.

The implementation also supports repeated evaluation across independently trained models. This is important for representation-level metrics because individual latent dimensions can vary across random seeds even when models implement similar functions. Accordingly, the experimental scripts preserve run-level results before computing aggregate statistics, allowing variability across independent training runs to be reported explicitly.

For the synthetic benchmark, the released code additionally supports the ground-truth analysis used in this paper. Since the generative factors of the synthetic dataset are known by construction, the learned bottleneck representations can be evaluated independently of InterpScore. This provides an external characterization of the representation and avoids incorporating the oracle information into the definition of the metric itself.

D.4 Metric Configuration

InterpScore exposes the weighting parameter α , which controls the relative contribution of the geometric and overlap/separation components of the metric. The implementation allows this parameter to be varied without retraining the evaluated model. Consequently, the same stored representations can be evaluated under different metric configurations.

This design also separates model-dependent computation from metric configuration. Once the required representations have been collected, different values of α can be evaluated without repeating model training or representation extraction. This property is useful both for sensitivity analysis and for future studies investigating alternative weightings of the constituent descriptors.

D.5 Extending InterpBench

The library is intended to support evaluation beyond the specific models and datasets considered in this paper. In a typical use case, a new model can be evaluated by exposing the desired representation layer, collecting its activations on an evaluation dataset, and passing the resulting representations to the InterpScore evaluation pipeline. The resulting score can then be compared across architectures, training procedures, bottleneck sizes, or other experimental conditions.

Because the metric is evaluated on representations rather than being coupled to a particular optimization objective, the same implementation can be used to study models trained under different supervised or adversarial training procedures. This property is exploited in the large-scale benchmark, where we evaluate models across multiple architectures, datasets, and training regimes.

D.6 Availability

The complete source code, experiment scripts, supplementary resources, and documentation are publicly available in the INTERPBENCH repository:

The repository also contains the scripts and resources required to reproduce the analyses reported in this paper. We recommend using the released implementation as the reference implementation of InterpScore for future comparisons and extensions.

Algorithm 1 Transition-Manifold Interpretability Benchmark

Require: Trained classifier f , dataset $\mathcal{D}$, layers $\{1, \dots, L\}$, attack parameters $(\epsilon, \alpha_{\text{PGD}}, T)$, PCA dimension k , reservoir capacity M , score parameters (α, Q) .

Ensure: Layer-wise scores $\{\mathcal{I}_l\}_{l=1}^L$, integrated score $\mathcal{I}_{\text{int}}$, and TTM.

1: Initialize streaming PCA estimators for all layers.

2: **Pass I: PCA estimation**

3: **for** each correctly classified $(x_n, y_n) \in \mathcal{D}$ **do**

4: Generate x_n^{adv} using PGD.

5: **if** $f(x_n^{\text{adv}}) \neq y_n$ **then**

6: **for** each layer l **do**

7: Extract and reduce $a_n^{(l)}$ to $z_n^{(l)}$.

8: **if** $z_n^{(l)}$ is finite **then**

9: Update the streaming PCA estimator for layer l .

10: **end if**

11: **end for**

12: **end if**

13: **end for**

14: Finalize PCA estimators to obtain $\{P_l, \mu_l\}_{l=1}^L$.

15: Initialize transition reservoirs $\mathcal{G}_{c_t, c_p}^{(l)}$.

16: **Pass II: transition geometry**

17: **for** each correctly classified $(x_n, y_n) \in \mathcal{D}$ **do**

18: Regenerate x_n^{adv} using PGD.

19: **if** $f(x_n^{\text{adv}}) \neq y_n$ **then**

20: Set $(c_t, c_p) = (y_n, f(x_n^{\text{adv}}))$.

21: **for** each layer l **do**

22: Extract and reduce $a_n^{(l)}$ to $z_n^{(l)}$.

23: **if** $z_n^{(l)}$ is finite **then**

24: Project:

$$r_n^{(l)} = P_l(z_n^{(l)} - \mu_l).$$

25: Insert $r_n^{(l)}$ into $\mathcal{G}_{c_t, c_p}^{(l)}$ subject to capacity M .

26: **end if**

27: **end for**

28: **end if**

29: **end for**

30: **for** each layer l **do**

31: Construct valid transition hulls $\mathcal{H}_{c_t, c_p}^{(l)}$.

32: Compute within-manifold descriptors and G_l .

33: Compute pairwise overlap O_l .

34: Compute pairwise separation S_l .

35: Compute

$$\mathcal{B}_l = G_l [1 + \alpha O_l (1 + Q S_l^{\text{norm}})].$$

36: Compute

$$\mathcal{I}_l = \frac{1}{1 + \mathcal{B}_l}.$$

37: Construct the class-conditioned TTM.

38: **end for**

39: Integrate $\{\mathcal{I}_l\}$ across normalized depth to obtain $\mathcal{I}_{\text{int}}$.

40: **return** $\{\mathcal{I}_l\}$, $\mathcal{I}_{\text{int}}$, and TTM.

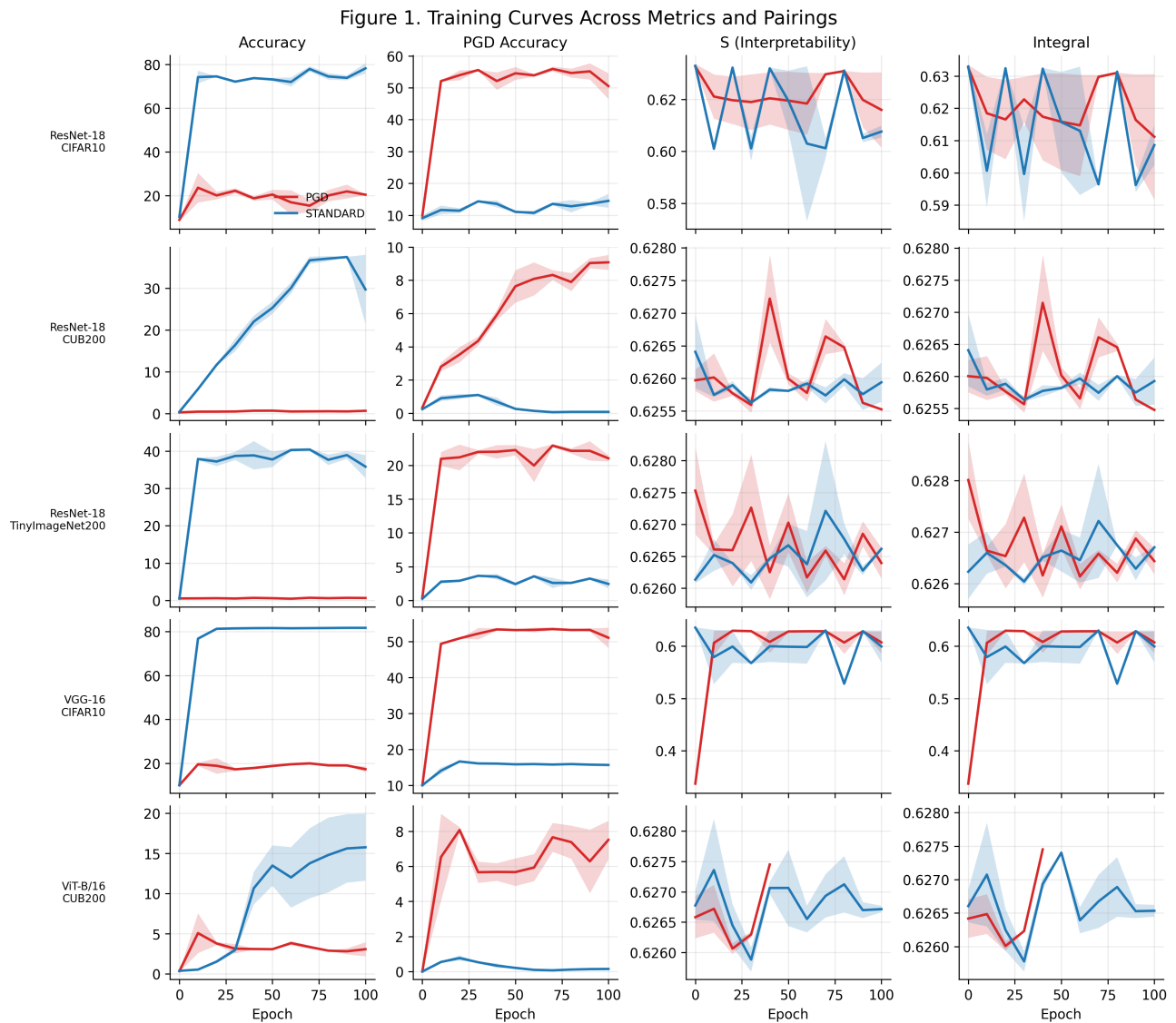


Figure 9: Complete training dynamics across the large-scale benchmark. Each panel reports task performance, adversarial performance, InterpScore, and InterpScore-Integral throughout training for an individual model-dataset configuration. These results provide the full set of trajectories underlying the representative stability analysis in the main paper.

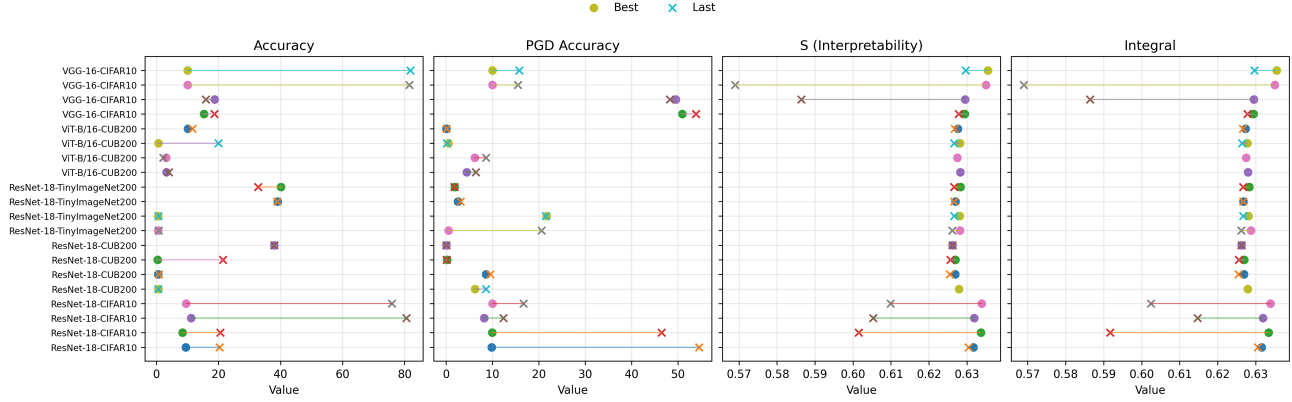


Figure 10: Comparison between best and final training checkpoints across the benchmark. Results are shown for both InterpScore and InterpScore-Integral. The comparison quantifies the sensitivity of the reported measurements to checkpoint selection.

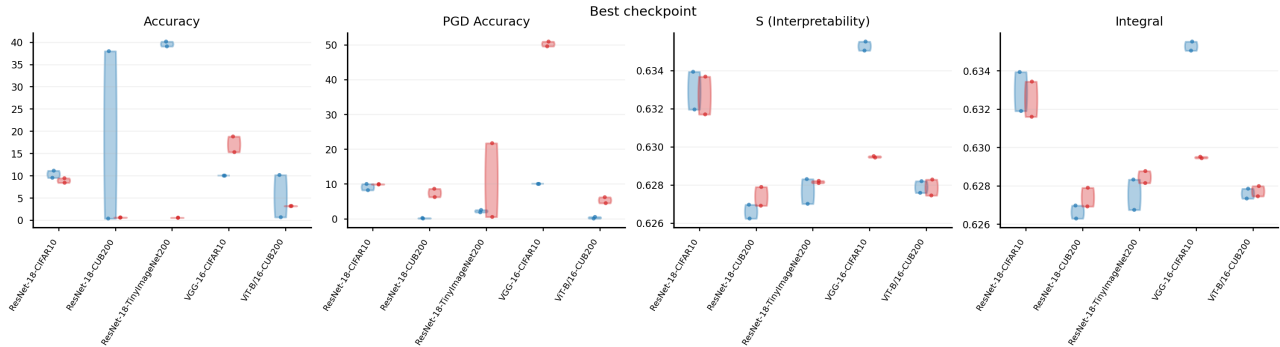


Figure 11: Distributions of InterpScore and InterpScore-Integral across evaluated model configurations. The distributions expose run-to-run and configuration-level variability that is summarized by the aggregate statistics reported in the main paper.

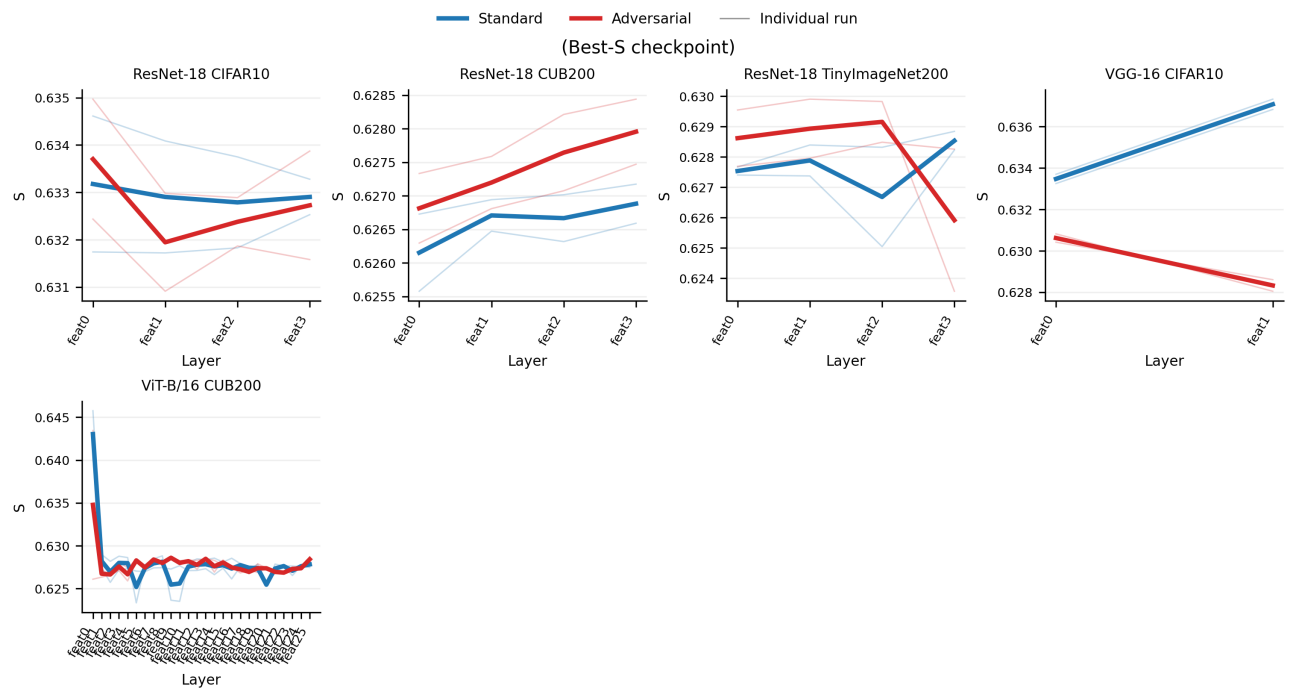
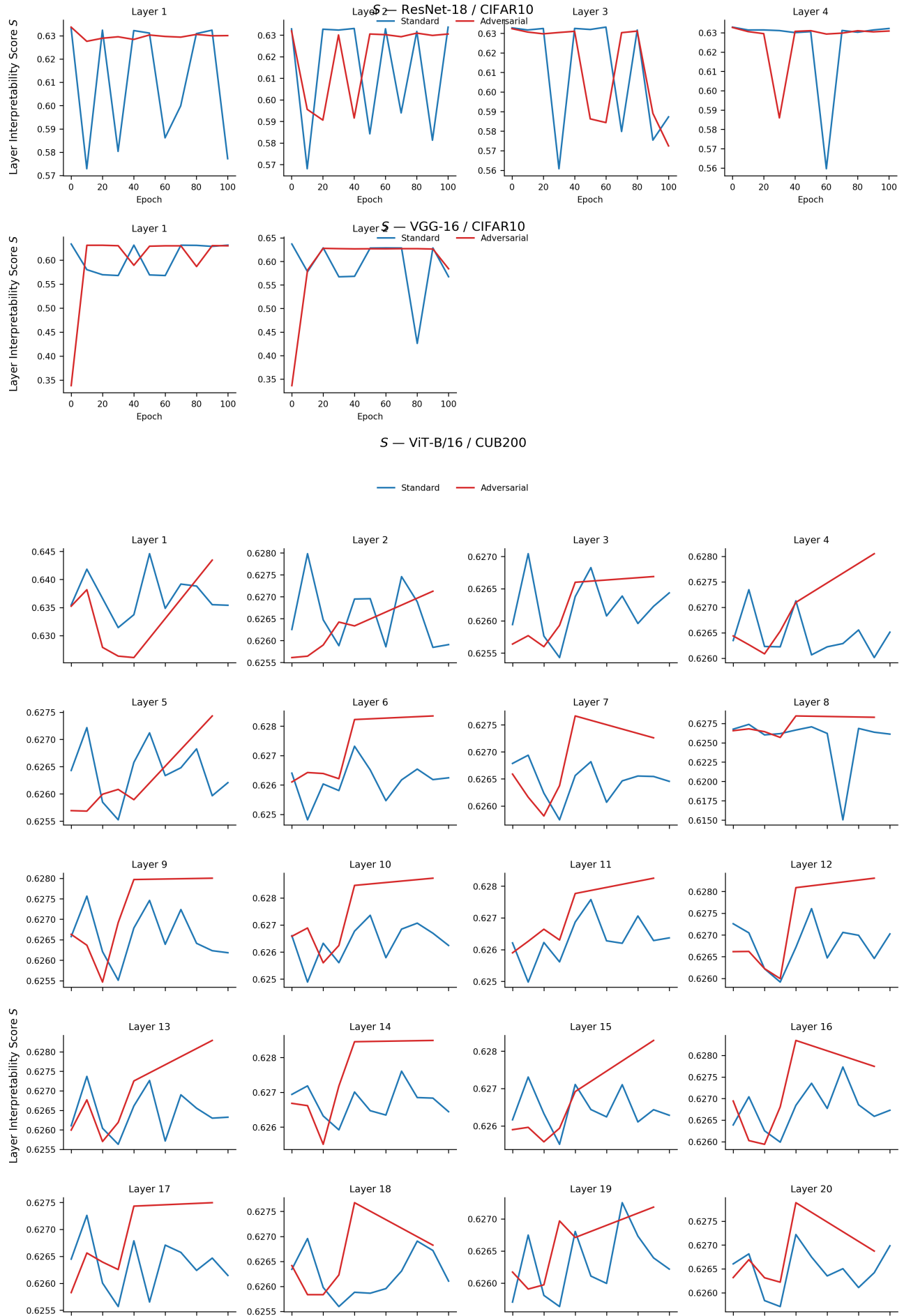


Figure 12: Layer-wise InterpScore across the evaluated networks. Scores are reported at the corresponding representation layers, revealing how interpretability varies with network depth.



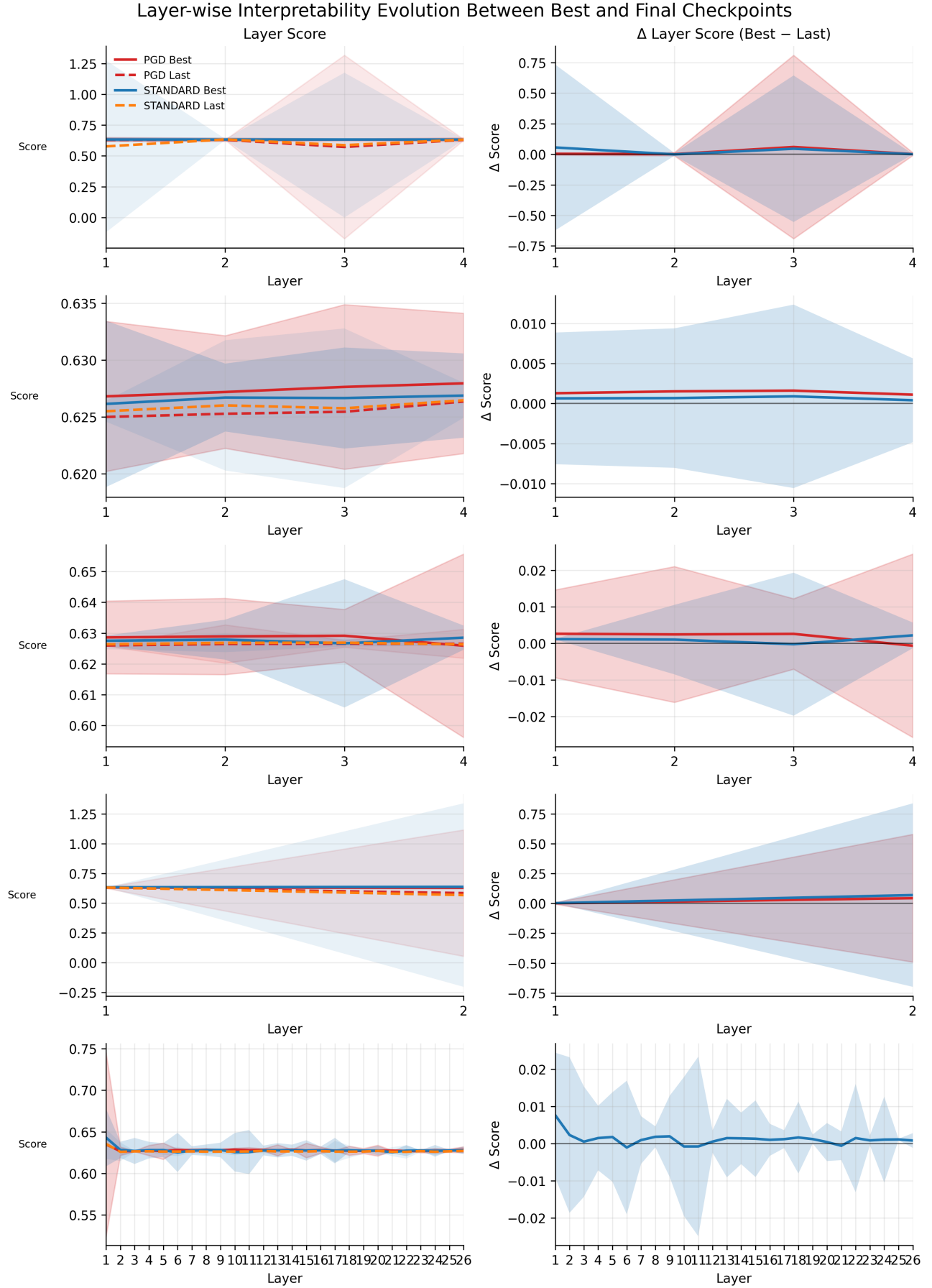


Figure 14: Individual-layer measurements of InterpScore across the evaluated models. These results provide the detailed layer-level measurements underlying the aggregate layer-wise analysis.

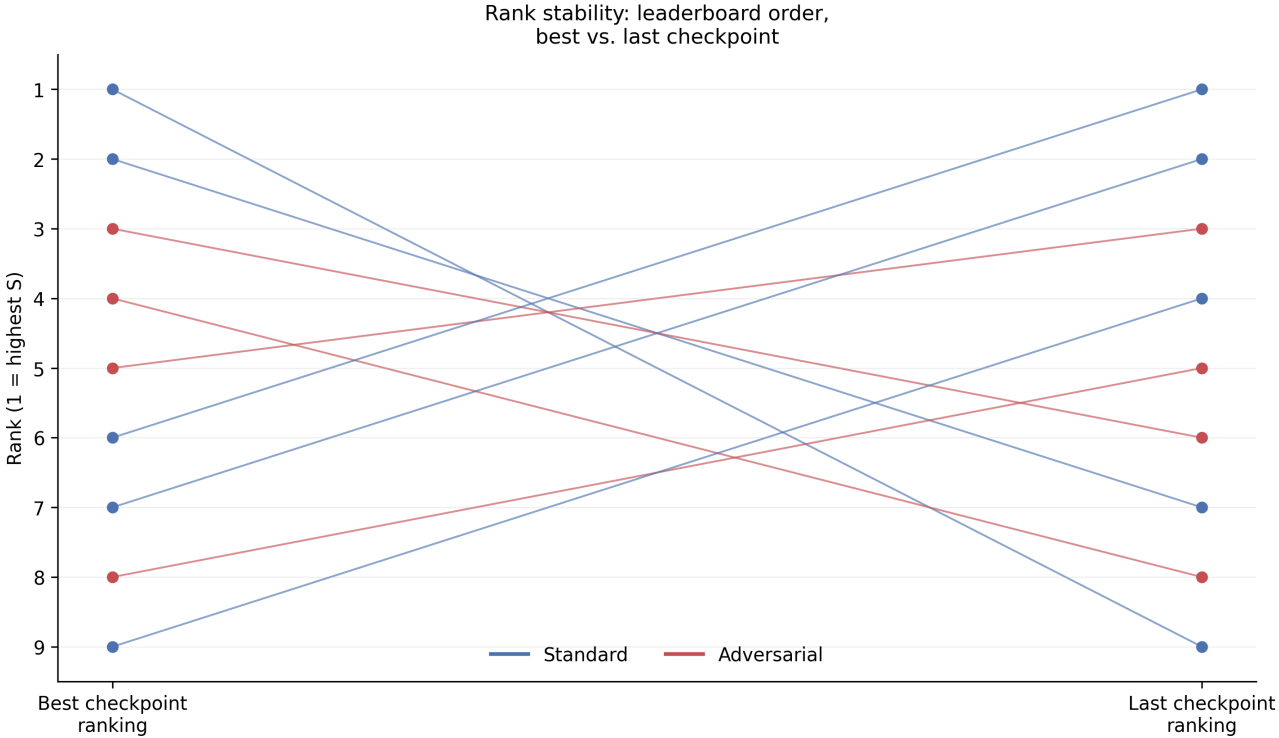


Figure 15: Rank stability of model comparisons under different evaluation settings. The analysis quantifies the consistency of model ordering across scoring choices and evaluation conditions.